

HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

GRADUATION THESIS

Optimizing Information Selection in Multimodal Models to Reduce Redundancy and Improve Performance

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Program: Data Science

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Department: Computer Engineering

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ABSTRACT

Vision-Language Models process images as sequences of visual tokens and use these tokens as evidence for language-based reasoning. Although retaining more visual tokens can preserve information, it also increases inference cost. Sparse visual-token selection reduces this cost by keeping only a subset of tokens, but aggressive pruning may discard useful evidence or retain multiple tokens that describe similar visual regions. This thesis studies this problem in image-based Vision-Language Models and focuses on improving SparseVLM-style token selection under constrained token budgets.

The thesis proposes a redundancy-aware selection strategy that augments text-guided token relevance with a diversity penalty. Instead of selecting visual tokens only by individual relevance scores, the proposed method selects from a high-relevance candidate pool while discouraging tokens that are too similar to already retained tokens. The method is evaluated against Dense / Vanilla inference, SparseVLM-Original, Threshold-Fixed, and Threshold-Adaptive settings on GQA, POPE, and a targeted failure-mining diagnostic set.

Experimental results show that the proposed method consistently improves over SparseVLM-Original on GQA and improves both POPE accuracy and F1 under the aggressive 64-token setting. On the failure-mining diagnostic set, it improves correctness from 70.00% to 75.00%, recovers 10 SparseVLM-Original failures, and introduces 5 regressions under the same aggressive setting. Auxiliary metrics also show that the proposed method reduces average pairwise similarity among selected tokens, supporting the intended redundancy-aware behavior. However, Threshold-Fixed remains a strong baseline in several broad benchmark settings, so the results do not support a claim of universal superiority. Overall, the thesis shows that balancing relevance and redundancy can improve SparseVLM-style visual-token selection, especially under aggressive pruning and targeted failure recovery.

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of answer correctness. 47

LIST OF ABBREVIATIONS

Abbreviation	Definition
AI	Artificial Intelligence
CLIP	Contrastive Language-Image Pretrain- ing
F1	F1-score
FN	False Negative
FP	False Positive
GQA	Graph Question Answering
LLM	Large Language Model
MLLM	Multimodal Large Language Model
MMR	Maximum Marginal Relevance
POPE	Polling-based Object Probing Evalua- tion
TN	True Negative
TP	True Positive
VLM	Vision-Language Model
VQA	Visual Question Answering

CHAPTER 1. Introduction

1.1 Research Background

Vision-Language Models (VLMs) combine visual understanding with natural-language processing, allowing models to answer questions, describe images, and follow multimodal instructions. In image-based VLMs, an input image is usually converted into a sequence of visual tokens. These tokens represent patches, regions, or learned visual features and are later used by the language component as visual evidence.

This token-based representation is flexible, but it can also be expensive. A single image may produce hundreds of visual tokens, and passing all of them through later model layers increases memory use and inference cost. As VLMs become larger and are used in more interactive applications, efficient visual-token processing becomes an important research problem.

1.2 Problem Statement

Sparse visual-token selection aims to reduce inference cost by keeping only a subset of visual tokens. A common approach is to assign each visual token an importance score and retain the highest-scoring tokens. SparseVLM follows this general direction by using text-guided attention information to identify visual tokens that are relevant to the input prompt.

However, selecting tokens only by individual relevance can be insufficient. Several retained tokens may still describe highly similar regions, while other useful evidence may be excluded. This issue becomes more severe when the token budget is aggressive, such as a nominal 64-token setting. Under such constraints, redundant selections consume limited capacity that could otherwise preserve small objects, attributes, relations, or other evidence needed for answering the question.

1.3 Motivation

The motivation of this thesis is that sparse token selection should consider both relevance and redundancy. A token should be useful for the current text prompt, but the selected set should also cover enough distinct evidence to support downstream reasoning. If the selected tokens are individually relevant but mutually redundant, the model may still miss the information required to answer correctly.

This thesis therefore investigates a redundancy-aware selection strategy for image-based VLM inference. The proposed direction does not aim to replace the relevance signal used by SparseVLM. Instead, it augments text-guided relevance with

an explicit diversity consideration so that selected visual tokens are less likely to duplicate each other under sparse budgets.

1.4 Research Objectives

The main objective of this thesis is to improve SparseVLM-style visual-token selection by reducing redundancy among selected visual tokens while preserving or improving answer-level performance under sparse inference.

Specifically, this thesis aims to:

1. Analyze redundancy in SparseVLM-style visual-token selection.
2. Develop a redundancy-aware visual-token selection method.
3. Evaluate whether redundancy-aware selection reduces redundancy and improves performance.
4. Analyze the accuracy-efficiency trade-off and diagnostic behavior of sparse selection methods.

1.5 Main Contributions

This thesis makes two main contributions.

1. **A redundancy-aware visual-token selection method.** This thesis develops a redundancy-aware selection strategy for SparseVLM-style image-based VLM inference. The method keeps SparseVLM’s text-guided relevance signal but modifies the final token-selection rule so that selected tokens are not only relevant to the prompt, but also less redundant with respect to already selected tokens.
2. **Evidence that redundancy-aware selection can reduce redundancy and improve performance.** The experiments show that Ours reduces average pair-wise similarity among selected visual tokens compared with SparseVLM-Original, supporting its intended redundancy-reduction behavior. It also improves answer-level performance over SparseVLM-Original in several sparse settings, especially under aggressive pruning on GQA, POPE, and the failure-mining diagnostic set.

1.6 Scope and Limitations

SparseVLM is designed for visual-token sparsification in Vision-Language Models and can be applied to both image and video understanding tasks [1]. However, this thesis focuses only on image-based VLM inference. Video inputs introduce additional challenges such as temporal redundancy, frame sampling, and long-context processing, which would require a different experimental setup. Therefore, video-

based tasks and datasets are outside the scope of this thesis.

Due to computational and scope limitations, this thesis does not evaluate all datasets used in the original SparseVLM paper. Instead, it focuses on two main image-based benchmarks: GQA and POPE. This choice is mainly due to the limited hardware setting, where experiments are conducted on a Kaggle environment with an NVIDIA P100 GPU, and to the image-only scope of the thesis. GQA is used to evaluate compositional visual question answering, including object recognition, attributes, relations, and multi-step reasoning [2]. POPE is used to evaluate object hallucination through yes/no object-existence questions [3]. In addition, a curated failure-mining set is used as a diagnostic stress test, but it is not treated as a representative benchmark distribution.

This thesis also does not exactly reproduce the full experimental setup of the original SparseVLM paper. Because of environment and hardware constraints, the experiments use LLaVA-v1.5-7B rather than the larger LLaVA-v1.5-13B variant, and all runs are conducted under the same local reproduction setting. For this reason, the Dense / Vanilla baseline is re-evaluated in the same environment instead of directly reusing full-token results from the original paper. This ensures that Dense / Vanilla, SparseVLM-Original, Ours, and the threshold-based baselines are compared under the same model backbone, dataset subsets, preprocessing choices, and evaluation scripts, although the resulting numbers should not be interpreted as directly comparable to results reported under different model sizes or hardware settings.

1.7 Thesis Organization

The remainder of this thesis is organized as follows. Chapter 2 reviews Vision-Language Models, visual-token representations, visual-token pruning, SparseVLM, and related work on redundancy in token selection. Chapter 3 describes the proposed redundancy-aware selection method and the threshold-based baselines. Chapter 4 presents the experimental setup, including datasets, compared methods, token budgets, evaluation metrics, and protocols. Chapter 5 reports the quantitative results, failure-driven analysis, visualization discussion, ablation study, and auxiliary token-selection metrics. Chapter 6 concludes the thesis, summarizes the main limitations, and outlines future research directions.

CHAPTER 2. Literature Review and Related Work

2.1 Vision-Language Models

Vision-Language Models (VLMs) are designed to process both visual and textual information, enabling tasks such as image captioning, image-text retrieval, visual question answering, and multimodal dialogue [4], [5]. Unlike traditional computer vision models that usually predict fixed labels from images, VLMs allow visual content to be interpreted and reasoned about through natural language.

In a typical image-based VLM, an image is first encoded into visual features or visual tokens. These visual representations are then aligned with, fused with, or passed into a language model to support language-based reasoning. For example, CLIP demonstrates that image and text representations can be effectively aligned through contrastive learning over large-scale image-text data [5]. More recent multimodal systems further connect visual encoders with large language models to support flexible question answering and instruction following over images.

Although VLMs have achieved strong performance on many multimodal tasks, processing visual information remains computationally expensive. Since images are often represented by many visual tokens, retaining all tokens may increase inference cost, while removing too many tokens may discard important visual evidence. Therefore, selecting useful visual information is an important problem in efficient image-based VLMs, which motivates the following discussion on visual token representations and token pruning.

2.2 Visual Token Representations in VLMs

In image-based Vision-Language Models, an image must first be converted into visual representations that can interact with language representations. These representations are often organized as visual tokens, where each token corresponds to a patch, region, or learned visual feature from the image. The language model then uses these tokens as visual evidence for tasks such as question answering, caption generation, or multimodal instruction following.

A common way to obtain visual tokens is to divide an image into patches and project each patch into an embedding vector. This idea is widely used in Transformer-based vision models such as Vision Transformer (ViT), where images are processed as sequences of patch embeddings [6]. This token-based representation makes it possible to connect visual information with language models, which also operate on token sequences.

The number of visual tokens is important because it affects both computational cost and information preservation. Keeping more tokens may preserve more visual details, but it also increases the cost of later processing. Keeping fewer tokens can improve efficiency, but may remove important evidence needed for downstream reasoning. Therefore, visual token representation naturally leads to the problem of visual token pruning and selection.

2.3 Visual Token Pruning and Token Selection

Since image-based VLMs often process images as sequences of visual tokens, using a large number of tokens can increase computational cost, especially when these tokens are passed into later Transformer layers or a language model. Visual token pruning and token selection address this problem by retaining only a smaller subset of useful tokens. A common strategy is to rank visual tokens by an importance score and keep the top-ranked ones. However, selecting tokens only by individual importance may not always preserve the best visual information: some selected tokens may be highly similar to each other, while other less obvious tokens may still be important for answering a specific question.

Previous studies have explored different ways to reduce visual tokens, including dynamic token pruning in DynamicViT [7] and token merging in ToMe [8]. In VLMs, this problem becomes especially important because removing visual tokens may also remove evidence needed for language-based reasoning. Recent work such as FastV has therefore investigated visual token pruning as a way to improve the efficiency of multimodal models [9].

Overall, visual token pruning introduces a trade-off between efficiency and information preservation. A good selection method should reduce unnecessary computation while still keeping the visual evidence required for downstream tasks. This motivates the need to consider not only token relevance, but also redundancy among the selected visual tokens.

2.4 SparseVLM: Visual Token Sparsification for Efficient Vision-Language Model Inference

SparseVLM is a visual token sparsification method proposed to improve the inference efficiency of Vision-Language Models (VLMs) [1]. Its main motivation is that visual tokens often introduce a large computational overhead, while not all visual tokens are equally useful for answering a given text query. Instead of processing all visual tokens throughout the model, SparseVLM aims to preserve the visual tokens that are most relevant to the input prompt and reduce less useful tokens during inference.

Unlike pruning methods that require additional training, SparseVLM is designed as a training-free and text-guided mechanism. It does not add new trainable modules or require fine-tuning. Instead, it uses attention information already available inside the VLM to estimate the relevance of visual tokens. The overall architecture of SparseVLM is illustrated in Figure 2.1.

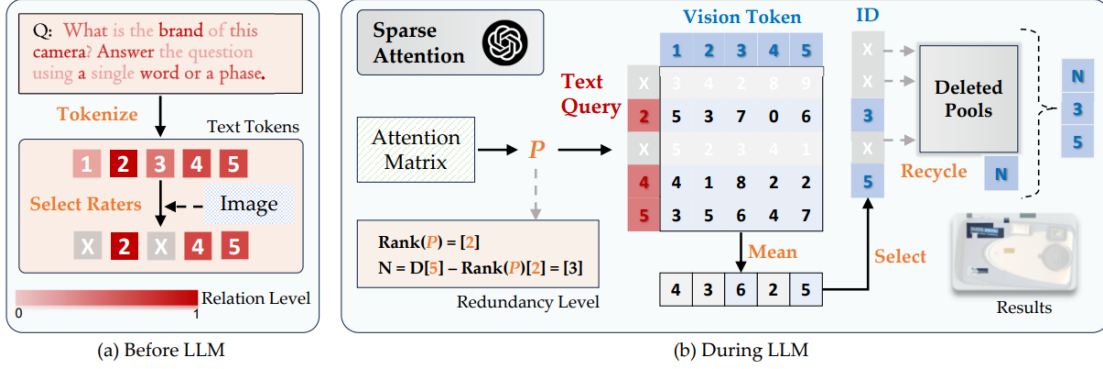


Figure 2.1: Overall architecture of SparseVLM. The method selects visually relevant text tokens before the LLM stage and uses them to guide visual-token sparsification during LLM inference. Pruned tokens can also be recycled into compact representations. Source: [1].

Figure 2.1 shows the main components of SparseVLM. The architecture can be summarized as follows:

1. **Visual encoding.** The input image is first processed by a visual encoder, which converts the image into a sequence of visual tokens.
2. **Text-token analysis.** The input prompt is represented as language tokens. SparseVLM identifies the text tokens that are most visually relevant to the image and uses them as text raters.
3. **Text-guided visual-token selection.** During LLM inference, the selected text raters guide the estimation of visual-token importance. Visual tokens that are more strongly related to the prompt are more likely to be retained.
4. **Layer-wise sparsification.** Instead of pruning all visual tokens only once at the input stage, SparseVLM performs sparsification at selected decoder layers. This allows the model to progressively reduce visual-token redundancy during inference.
5. **Token recycling.** Pruned visual information is not always discarded completely. SparseVLM compresses part of the removed visual information into compact recycled tokens, helping reduce information loss while still improving efficiency.

The procedure of SparseVLM consists of four main steps:

1. Identifying visually relevant text tokens from the input prompt.
2. Scoring visual tokens based on their relevance to the selected text tokens.
3. Applying visual-token sparsification across multiple decoder layers.
4. Recycling part of the pruned visual information to reduce information loss.

This design makes SparseVLM different from text-agnostic token pruning methods. Since the selection process is guided by the input prompt, the retained visual tokens can vary depending on the question being asked. For example, two different questions about the same image may require different visual evidence, and text-guided sparsification is intended to preserve the visual regions most relevant to each question.

Figure 2.2 provides a qualitative example of SparseVLM’s token selection behavior.



Figure 2.2: Qualitative demonstration of SparseVLM. The retained visual tokens become progressively focused on regions that are relevant to the input question. Source: [1].

Figure 2.2 illustrates how SparseVLM progressively concentrates visual tokens around regions that are more relevant to the input question. In the earlier layers, the retained tokens may still cover a broader part of the image. In later layers, the selected tokens become more focused on the visual evidence needed for answering the question. This example highlights the intuition behind text-guided visual token sparsification: many image patches may be unnecessary for a specific query, and reducing them can improve efficiency while preserving task-relevant information.

In this thesis, SparseVLM is used as the main sparse baseline because it directly addresses efficient visual token selection in VLMs. However, although SparseVLM is text-guided and adaptive across layers, its selection mechanism may still retain

redundant visual tokens or miss visual evidence that is important for answering specific questions, especially under aggressive token budgets. This limitation motivates the proposed redundancy-aware selection method, which aims to balance token relevance with diversity among selected visual tokens.

2.5 Redundancy in Visual Token Selection

Visual token pruning is not only about removing unimportant tokens, but also about avoiding redundancy among the retained tokens. In image-based VLMs, nearby patches or visually similar regions may produce tokens that contain overlapping information. This intuition is also related to token-merging methods such as ToMe, where similar tokens are combined to reduce computation in Transformer models [8]. As a result, a method that selects tokens only by individual importance may still keep several tokens describing similar visual content, especially under small token budgets. If many retained tokens focus on the same dominant region, the model may have less capacity to preserve other visual evidence needed for reasoning, such as small objects, attributes, or spatial relations.

In this thesis, redundancy is approximated using the average pairwise cosine similarity between selected visual tokens:

$$\text{Sim}_{\text{avg}}(\mathcal{S}) = \frac{2}{|\mathcal{S}|(|\mathcal{S}| - 1)} \sum_{i < j, v_i, v_j \in \mathcal{S}} \text{sim}(v_i, v_j),$$

where $\mathcal{S}$ is the selected token set and $\text{sim}(v_i, v_j)$ is computed using cosine similarity. A higher value suggests that the selected tokens may contain more overlapping information in the model’s representation space. However, this metric should be interpreted as a proxy for redundancy, not as a complete measure of visual evidence quality.

Reducing redundancy alone is not sufficient. If a method focuses too strongly on diversity, it may select visually different but task-irrelevant tokens. Therefore, the central challenge is to balance relevance and diversity. This idea is closely related to maximal marginal relevance, which selects items by considering both their relevance and their novelty with respect to already selected items [10]. This thesis is motivated by the same general trade-off and proposes a redundancy-aware selection method that aims to preserve task-relevant evidence while reducing unnecessary similarity among retained visual tokens.

2.6 Related Work Summary and Research Gap

From VLMs to visual-token cost. The previous sections reviewed the main background for this thesis. Image-based Vision-Language Models (VLMs) pro-

cess visual and textual information jointly, enabling tasks such as visual question answering, image-text retrieval, captioning, and multimodal dialogue [4], [5]. In these models, images are commonly represented as visual tokens or patch-level embeddings, following the broader token-based design used in Transformer-based vision models such as Vision Transformer [6]. This representation makes it possible to connect visual information with language models, but it also introduces a computational challenge: an image may produce many visual tokens, and processing all of them throughout the model can be expensive.

From general token reduction to VLM token pruning. Prior work on token reduction shows that not all visual tokens need to be processed equally. DynamicViT reduces computation by dynamically pruning less informative tokens in Vision Transformers [7], while ToMe reduces token redundancy by merging similar tokens rather than discarding them directly [8]. These works demonstrate that token reduction is a promising strategy for improving efficiency. However, they are mainly designed for visual recognition or general Transformer efficiency, where the objective is not necessarily conditioned on a language query. In image-based VLMs, the importance of a visual token depends not only on the image itself, but also on the specific question or prompt. Therefore, token reduction in VLMs must preserve visual evidence that is relevant to language-based reasoning.

From VLM pruning to SparseVLM. More recent work has studied visual-token reduction directly in VLMs. FastV investigates visual token pruning for efficient VLM inference and shows that many visual tokens can be reduced while maintaining much of the model’s performance [9]. SparseVLM is especially relevant to this thesis because it provides a training-free, text-guided visual token sparsification method for VLM inference [1]. By using visually relevant text tokens as raters, SparseVLM selects visual tokens that are more related to the input prompt. It also applies layer-wise sparsification and token recycling to reduce inference cost while preserving part of the pruned information.

Research gap. Based on the reviewed literature, the research gap can be stated directly in three parts:

1. **General visual token reduction methods do not fully address prompt-conditioned visual reasoning.** DynamicViT and ToMe show that reducing or merging visual tokens can improve efficiency [7], [8]. However, these methods are mainly designed for vision models or general Transformer efficiency. In image-based VLMs, the importance of a visual token depends not only on

the image, but also on the specific text prompt or question. Therefore, VLM token selection must preserve visual evidence that is useful for language-based reasoning, not only reduce visual computation.

2. **SparseVLM provides prompt-aware selection, but its retained set is still mainly determined by individual relevance.** SparseVLM is highly relevant to this thesis because it performs training-free, text-guided visual token sparsification for VLM inference [1]. Its selection mechanism uses the input text to estimate which visual tokens are important. However, selecting tokens according to individual relevance does not explicitly consider the relationship among the retained tokens. Several selected tokens may still describe similar regions or overlapping visual evidence. Under aggressive sparse budgets, this can reduce the capacity available for other useful evidence, such as small objects, attributes, or spatial relations.
3. **The relevance–redundancy trade-off is well motivated, but remains underexplored in the SparseVLM-style selection setting.** The idea of balancing relevance and redundancy is well established in maximal marginal relevance, where selected items should be relevant while also adding novel information with respect to previously selected items [10]. Similarity-based token merging also reflects the intuition that highly similar tokens may contain overlapping information [8]. However, within the reviewed VLM token-pruning literature, this relevance–redundancy trade-off is not the central focus of the sparse visual-token selection rule. In particular, SparseVLM provides a strong text-guided baseline, but its direct relevance-based selection step does not explicitly penalize redundancy among retained visual tokens.

This thesis addresses the above gap by proposing and evaluating a redundancy-aware token selection strategy for image-based VLM inference. Instead of replacing SparseVLM’s text-guided relevance signal, the proposed method keeps the same general SparseVLM pipeline and modifies the final selection rule. The goal is to retain visual tokens that are both relevant to the input prompt and less redundant with respect to the tokens already selected.

CHAPTER 3. Proposed Method

3.1 Overview

This chapter presents the visual token selection methods used in this thesis. It first summarizes the baseline selection behavior of SparseVLM, which serves as the main reference method. It then introduces the proposed redundancy-aware selection method, which aims to balance token relevance and diversity during sparse visual token selection.

In addition to the proposed method, this chapter also describes two threshold-based baselines. These baselines provide simpler redundancy-aware alternatives and are included to make the comparison more informative. The chapter ends by summarizing the main methodological differences among SparseVLM-Original, the proposed method, Threshold-Fixed, and Threshold-Adaptive.

3.2 Baseline Token Selection in SparseVLM

As reviewed in Section 2.4, SparseVLM is a text-guided visual token sparsification method for efficient VLM inference [1]. In this thesis, SparseVLM-Original is used as the main sparse baseline. In the implementation used for the experiments, this baseline corresponds to the `topk` selection mode, where visual tokens are ranked by text-guided attention scores and the highest-scoring tokens are retained according to a layer-wise keep budget.

Baseline procedure. The baseline procedure follows the main text-guided sparsification mechanism introduced in SparseVLM, while the experiments in this thesis evaluate it through the fixed-budget `topk` implementation branch. The procedure can be described as follows:

1. **Prepare visual and textual representations.** The input image is encoded into visual tokens, and the input prompt is represented as text tokens. SparseVLM then reuses attention information from the VLM decoder instead of adding a separate trainable pruning module.
2. **Select visually relevant text raters.** SparseVLM first identifies text tokens that are strongly related to the visual input. These selected text tokens are used as visual-relevant raters, because not every prompt token is equally useful for judging the importance of visual tokens.
3. **Estimate visual-token significance using attention.** After the visually relevant text raters are selected, SparseVLM uses the attention relationship be-

tween text tokens and visual tokens to estimate the significance of each visual token. The paper first defines the decoder self-attention matrix as:

$$A = \text{Attention}(Q, K) = \text{Softmax} \left(\frac{QK^\top}{\sqrt{D}} \right),$$

where Q and K are the query and key matrices, and D is the dimensionality used for scaling.

SparseVLM then extracts the language-to-vision part of the attention matrix as the sparsification priority matrix:

$$P = A_{i,j}, \quad (i, j) \in \{\mathcal{L}, \mathcal{I}\},$$

where $\mathcal{L}$ denotes the language instruction token set and $\mathcal{I}$ denotes the image token set.

The visual-token significance is estimated by averaging the priority scores over the text dimension:

$$\tilde{\mathbf{p}} = [\tilde{p}_1, \tilde{p}_2, \dots, \tilde{p}_{L_v}] = \frac{1}{L_t} \sum_{i=1}^{L_t} P_i.$$

A larger value in $\tilde{\mathbf{p}}$ indicates that the corresponding visual token is considered more significant for text-guided sparsification.

4. **Apply sparse visual-token selection.** In the original SparseVLM paper, the sparsification level is adaptively determined using a rank-based strategy:

$$N = \lambda \times (L_v - \text{rank}(P)),$$

where N is the number of visual tokens to be removed, L_v is the number of visual tokens, and λ is a scaling factor.

In this thesis, however, SparseVLM-Original is evaluated through the fixed-budget `topk` branch of the implementation. Therefore, after obtaining text-guided visual-token scores, the baseline keeps the highest-scoring visual tokens according to a fixed layer-wise keep schedule. The nominal 128-token setting maps to [303, 110, 36], while the nominal 64-token setting maps to [66, 30, 17]. These values are target budgets for selected current visual tokens at sparse layers, not exact final visual-token sequence lengths.

5. **Recycle part of the pruned visual information.** SparseVLM also includes

token recycling to reduce information loss. Instead of discarding all pruned visual tokens completely, part of the removed visual information can be clustered and compressed into compact recycled tokens, which are inserted back into the sequence.

Limitations. Although SparseVLM uses text guidance and rank-based sparsification to improve efficiency, the fixed-budget topk baseline used in this thesis still ranks and retains visual tokens primarily according to individual text-guided significance. It does not explicitly evaluate whether the retained visual tokens are mutually redundant or whether a newly selected token adds information that is different from the tokens already selected.

This limitation becomes more important under aggressive nominal sparse budgets, especially in the 64-token setting. When multiple selected tokens encode overlapping visual evidence, fewer positions remain available for other potentially useful evidence, such as small objects, attributes, or spatial relations. As a result, the retained set may contain individually relevant tokens but may still be inefficient as a set. This motivates the redundancy-aware selection strategy introduced in the next section.

3.3 Proposed Redundancy-Aware Selection (Ours)

Overall idea. The proposed method modifies the final visual-token selection step of SparseVLM-Original. In the rest of this thesis, this method is referred to as Ours. The main idea is to keep the original text-guided relevance signal from SparseVLM, but change how the retained visual-token set is formed. Instead of selecting tokens only because they have high individual relevance scores, Ours also considers whether a candidate token is redundant with tokens that have already been selected.

Architecturally, Ours does not introduce a new visual encoder, language model, or trainable pruning module. It is implemented as a replacement for the direct topk selection rule inside the SparseVLM sparse inference pipeline. The visual encoder, text-guided attention scoring, sparse layers, layer-wise keep schedules, and token recycling mechanism remain unchanged. Therefore, the difference between SparseVLM-Original and Ours lies mainly in the selection rule used to choose the retained original visual tokens.

Implementation setting. In the implementation, Ours corresponds to the `mmr` selection branch. The term MMR is used here in the sense of maximum marginal relevance: a selected token should be relevant to the input prompt, but it should

also provide information that is not too similar to the tokens already selected.

At a sparse layer ℓ , the method starts from the same text-guided relevance scores used by SparseVLM-Original. Let $s_i^{(\ell)}$ denote the relevance score of visual token i at layer ℓ , and let k_ℓ denote the target number of selected current visual tokens at that sparse layer. These scores are not replaced by a new scoring model. Instead, Ours uses them as the relevance component in a relevance–redundancy selection rule.

Candidate-pool construction. The method first constructs a candidate pool from the most relevant visual tokens. Rather than applying redundancy-aware selection over all visual tokens, Ours restricts the search to a larger pool of high-relevance candidates:

$$\mathcal{C}^{(\ell)} = \text{TopK} \left(\{s_i^{(\ell)}\}_{i=1}^{N_\ell}, \min(\alpha k_\ell, N_\ell) \right),$$

where N_ℓ is the number of current visual tokens at layer ℓ , k_ℓ is the target keep budget, and α is the candidate-pool factor. In the implementation, α corresponds to `candidate_pool_factor`.

This candidate pool keeps the method focused on tokens that are already relevant to the prompt, while still giving the selection rule more choices than direct TopK selection.

Redundancy measurement. Ours measures redundancy using cosine similarity between current-layer visual-token hidden states. For two visual tokens i and j , the similarity is computed as:

$$\text{sim} \left(h_i^{(\ell)}, h_j^{(\ell)} \right) = \frac{\left(h_i^{(\ell)} \right)^\top h_j^{(\ell)}}{\left\| h_i^{(\ell)} \right\|_2 \left\| h_j^{(\ell)} \right\|_2},$$

where $h_i^{(\ell)}$ and $h_j^{(\ell)}$ are the current-layer hidden-state representations of the two visual tokens. This similarity is computed from the model’s hidden representations, not from raw image patches.

For a candidate token i , its redundancy with respect to the already selected set $\mathcal{S}^{(\ell)}$ is defined using the maximum similarity to any selected token:

$$r_i^{(\ell)} = \max_{j \in \mathcal{S}^{(\ell)}} \max \left(0, \text{sim} \left(h_i^{(\ell)}, h_j^{(\ell)} \right) \right).$$

The maximum operation reflects the intuition that a candidate token is redundant

if it is highly similar to at least one token that has already been retained. Negative similarities are clipped to zero before being used as redundancy penalties.

Relevance–redundancy scoring. After the candidate pool is constructed, Ours selects tokens greedily. The first selected token is chosen mainly according to relevance. After that, each remaining candidate is scored by combining normalized relevance with a redundancy penalty:

$$g_i^{(\ell)} = \lambda \tilde{s}_i^{(\ell)} - (1 - \lambda) r_i^{(\ell)},$$

where $\tilde{s}_i^{(\ell)}$ is the normalized relevance score, $r_i^{(\ell)}$ is the redundancy score, and $\lambda \in [0, 1]$ controls the trade-off between relevance and redundancy.

A larger λ makes the method behave more similarly to relevance-based TopK selection, because more weight is placed on the original SparseVLM relevance score. A smaller λ gives more influence to the redundancy penalty, encouraging the method to select tokens that are less similar to those already retained.

Greedy selection process. The selected set is built iteratively. At each step, Ours chooses the candidate with the highest relevance–redundancy score.

The selected token is then added to $\mathcal{S}^{(\ell)}$, and the redundancy scores of the remaining candidates are updated with respect to the new selected set. Already selected candidates are masked out so that the same token cannot be selected more than once. This process continues until the target number of selected current visual tokens is reached, subject to the available candidate pool and current visual-token count.

Interaction with the SparseVLM pipeline. After the redundancy-aware selection mask is produced, the remaining SparseVLM sparse inference pipeline is unchanged. Prompt tokens and question tokens are preserved. Token recycling/compression is still applied when pruned visual tokens are available, so part of the removed visual information may be compressed and reinserted into the sequence.

The purpose of Ours is therefore not to maximize diversity alone. A visually different token may still be unhelpful if it is irrelevant to the input question. Instead, Ours is designed to preserve the text-guided relevance signal from SparseVLM while discouraging the selection of visual tokens that are too similar to tokens already retained.

3.4 Threshold-Based Baselines

In addition to SparseVLM-Original and Ours, this thesis includes threshold-based baselines to examine whether redundancy-aware selection can also be implemented through a simpler hard-filtering rule. These baselines use the same SparseVLM sparse inference pipeline and the same text-guided relevance signal, but they differ in how they handle similarity among selected visual tokens.

The main idea of threshold filtering is to reject a candidate token when it is too similar to tokens that have already been selected. This provides a direct and interpretable way to reduce redundancy. Two threshold-based variants are included for different comparison purposes:

1. **Threshold-Fixed** keeps the same fixed selected-token budget as SparseVLM-Original and Ours. This makes it directly comparable to Ours under the same nominal sparse-budget settings. Since a strict threshold may reject too many tokens, Threshold-Fixed uses a backfill procedure to maintain the fixed budget.
2. **Threshold-Adaptive** does not force the selected-token count to match a fixed budget. Instead, it allows the number of retained tokens to vary according to the similarity threshold. This makes it useful for studying the token–performance trade-off, but less directly comparable to fixed-budget methods such as SparseVLM-Original, Ours, and Threshold-Fixed.

3.4.1 Threshold-Fixed

Rationale. Threshold-Fixed is included as a fixed-budget redundancy-aware baseline. Its purpose is to test whether a simple hard-threshold rule can compete with Ours when both methods retain the same target number of selected visual tokens.

Shared components with SparseVLM and Ours. Threshold-Fixed keeps the same overall SparseVLM sparse inference pipeline. It uses the same visual encoder, text-guided relevance scores, sparse layers, nominal keep schedules, and token recycling mechanism as SparseVLM-Original. It also reuses the same relevance-based candidate-pool construction and hidden-state cosine similarity computation used in Ours, as described in Section 3.3.

Therefore, this subsection focuses only on the parts that differ from SparseVLM-Original and Ours.

Hard-threshold selection rule. After the shared candidate pool has been constructed, Threshold-Fixed differs from Ours in how it selects tokens from that pool.

Ours combines relevance and redundancy through a continuous score, whereas Threshold-Fixed uses a binary acceptance rule based on a fixed similarity threshold τ .

For a candidate token i , its maximum similarity to the already selected set $\mathcal{S}^{(\ell)}$ is computed as:

$$d_i^{(\ell)} = \max_{j \in \mathcal{S}^{(\ell)}} \text{sim} \left(h_i^{(\ell)}, h_j^{(\ell)} \right),$$

where $h_i^{(\ell)}$ and $h_j^{(\ell)}$ are current-layer hidden-state representations of visual tokens. The cosine similarity function is the same as the one used in Ours.

A candidate token is accepted if:

$$d_i^{(\ell)} \leq \tau.$$

If no token has been selected yet, the first candidate is accepted. Candidates are scanned in descending order of text-guided relevance. Therefore, Threshold-Fixed still prioritizes relevant tokens, but it rejects a candidate when the candidate is too similar to tokens that have already been retained.

Backfill for fixed-budget selection. Threshold-Fixed is designed as a fixed-budget baseline. However, a strict similarity threshold may reject too many candidates, leaving fewer selected visual tokens than the target keep budget k_ℓ . To avoid changing the selected-token budget, Threshold-Fixed uses a backfill procedure.

If the threshold-filtered selected set is smaller than k_ℓ , the remaining positions are filled using the following priority:

1. outside-pool tokens that still satisfy the similarity threshold;
2. outside-pool tokens ranked by relevance, ignoring the threshold;
3. any remaining unselected tokens ranked by relevance.

This backfill step is the key difference between Threshold-Fixed and Threshold-Adaptive. Threshold-Fixed tries to preserve the fixed selected-token budget, while Threshold-Adaptive allows the number of selected tokens to vary according to the threshold.

Interaction with the SparseVLM pipeline. After the threshold-based selection mask is produced, the remaining SparseVLM inference pipeline is unchanged. Prompt tokens and question tokens are preserved, and token recycling/compression is still applied when pruned visual tokens are available.

Threshold-Fixed is useful because it provides a simple and interpretable redundancy-aware baseline. However, the hard-threshold rule is less flexible than the continuous relevance–redundancy scoring used in Ours. If the threshold is too strict, the method may reject relevant tokens. If the threshold is too loose, the method may behave similarly to relevance-based selection and retain redundant tokens. In addition, the backfill step can reintroduce tokens that do not satisfy the threshold when necessary to maintain the fixed budget.

3.4.2 Threshold-Adaptive

Rationale. Threshold-Adaptive is included as a variable-token redundancy-aware baseline. Its purpose is to observe how hard-threshold filtering behaves when the number of retained visual tokens is allowed to change according to the similarity threshold.

Variable-token behavior. Threshold-Adaptive reuses the same SparseVLM sparse inference pipeline, text-guided relevance scores, relevance-based candidate pool, hidden-state cosine similarity, and hard-threshold acceptance rule described in Threshold-Fixed. The key difference is that Threshold-Adaptive does not use the backfill procedure.

After the candidate pool has been scanned, the selected set contains only the candidates that satisfy the similarity threshold. If many candidates are similar to tokens that have already been selected, more candidates are rejected and the retained-token count becomes smaller. If the candidate pool contains more diverse candidates, more tokens may pass the threshold. Therefore, the number of retained visual tokens can vary across examples and layers.

Interpretation. Because Threshold-Adaptive does not enforce a fixed selected-token budget, it is less directly comparable to fixed-budget methods such as SparseVLM-Original, Ours, and Threshold-Fixed. Instead, it is mainly used as a token–performance trade-off baseline. It helps show how a hard redundancy threshold behaves when the model is allowed to retain a variable number of visual tokens, rather than being forced to match a fixed nominal budget.

3.5 Methodological Differences

This section summarizes the main methodological differences among the token selection methods used in this thesis. All methods start from the same SparseVLM text-guided relevance score, but they differ in how the final selected visual-token set is constructed. They also share the same SparseVLM sparse inference pipeline after selection, including token recycling/compression when pruned to-

kens are available.

It is important to note that the budget descriptions in this thesis refer to selected current visual tokens at each sparse layer. They should not be interpreted as exact final visual-token sequence lengths, because token recycling/compression may add compressed tokens after the selection mask is produced. They should also not be interpreted as exact counts of original image-patch tokens, since later sparse layers may operate on visual-token sequences that already include compressed representations.

Table 3.1 summarizes the comparison.

Table 3.1: Methodological comparison of the visual token selection methods used in this thesis.

Method	Selection style	Redundancy handling	handling	Budget behavior
SparseVLM-Original	Direct TopK over text-guided relevance scores	No explicit redundancy penalty	redundancy	Fixed selected current visual-token budget, capped by the current visual-token count
Ours	Greedy selection from a relevance-based candidate pool	Cosine-similarity redundancy penalty controlled by λ	penalty	Fixed selected current visual-token budget, capped by the current visual-token count
Threshold-Fixed	Relevance-ordered threshold filtering with backfill	Hard threshold on maximum cosine similarity to selected tokens	on cosine selected	Fixed selected current visual-token budget with backfill, capped by the current visual-token count
Threshold-Adaptive	Relevance-ordered threshold filtering without backfill	Hard threshold on maximum cosine similarity to selected tokens	on cosine selected	Variable selected current-token count; no fixed-budget enforcement

The table highlights the main methodological differences among the compared sparse selection methods. SparseVLM-Original serves as the relevance-based TopK baseline. Ours modifies this selection rule by adding a continuous redundancy penalty while keeping the rest of the SparseVLM pipeline unchanged. Threshold-Fixed also introduces redundancy filtering, but it uses a hard similarity threshold and a backfill procedure to preserve the fixed selected-token budget. Threshold-Adaptive uses the same threshold-based idea without backfill, so its retained-token count can vary across examples and layers.

This distinction is important for the experiments in later sections. SparseVLM-Original, Ours, and Threshold-Fixed are fixed-budget methods and can therefore

be compared under the same nominal sparse-budget settings. Threshold-Adaptive is instead interpreted as a variable-token trade-off baseline rather than a directly fixed-budget competitor.

CHAPTER 4. Experimental Setup

4.1 Datasets

This thesis evaluates visual-token selection using two benchmark evaluation settings and one targeted diagnostic set. GQA and POPE are used for the main quantitative benchmark evaluation, while the failure-mining set is used for failure-driven analysis and qualitative inspection. In addition, deterministic subsets of GQA and POPE are used for ablation studies and auxiliary selected-token metrics. This separation is important because the benchmark datasets measure overall task performance, whereas the diagnostic and ablation subsets are used to analyze the behavior of the selection mechanisms.

4.1.1 GQA

GQA is used as the main visual question answering benchmark in this thesis. It contains questions that require object recognition, attribute understanding, spatial relations, and compositional reasoning over images [2]. These properties make it suitable for evaluating whether sparse visual-token selection preserves the visual evidence needed for image-conditioned answering.

GQA is designed to evaluate compositional visual reasoning over real-world images. Unlike simpler visual question answering datasets that may focus mainly on object recognition, GQA questions often require the model to identify objects, inspect attributes, compare entities, follow spatial or semantic relations, and combine multiple reasoning steps. Each image is associated with a scene graph containing objects, attributes, and relations, while each question is associated with a functional program that represents its reasoning structure [2].

Table 4.1: Key characteristics of the GQA dataset.

Characteristic	Description
Number of images	113,018 real-world images
Number of questions	22,669,678 questions
Vocabulary size	3,097 words
Answer space	1,878 possible answers
Question structure	Includes binary, open-ended, choice, logical, and comparison questions
Semantic focus	Includes object, attribute, relation, category, and global-scene questions
Reasoning complexity	Most questions require multiple reasoning steps, making the dataset suitable for compositional visual reasoning

These characteristics make GQA suitable for this thesis because visual-token selection must preserve not only salient objects, but also smaller details and spatially relevant evidence. If aggressive pruning removes tokens corresponding to attributes, relations, or less dominant objects, the model may lose information needed to answer GQA questions correctly.



- A1. Is the **tray** on top of the **table** black or light brown? light brown
A2. Are the **napkin** and the **cup** the same color? yes
A3. Is the small **table** both oval and wooden? yes
A4. Is there any **fruit** to the left of the **tray** the **cup** is on top of? yes
A5. Are there any **cups** to the left of the **tray** on top of the **table**? no
B1. What is the brown **animal** sitting inside of? **box**
B2. What is the large **container** made of? cardboard
B3. What **animal** is in the **box**? **bear**
B4. Is there a **bag** to the right of the green **door**? no
B5. Is there a **box** inside the plastic **bag**? no

Figure 4.1: Example image-question-answer pairs from the GQA dataset.

Figure 4.1 illustrates the type of visual reasoning required in GQA. Each image is associated with multiple questions that ask about different aspects of the scene, such as object existence, object attributes, color, material, spatial relations, and containment relations. Some questions are binary yes/no questions, while others require short open-ended answers. This structure makes GQA suitable for evaluating image-based VLMs because the model must not only recognize individual objects, but also preserve enough visual evidence to reason about relationships between objects. In the context of this thesis, such questions are especially relevant because aggressive visual-token pruning may remove small or spatially important evidence needed to answer correctly.

4.1.2 POPE

POPE is used to evaluate object hallucination in image-based Vision-Language Models. Instead of asking open-ended questions, POPE formulates object hallu-

cination as a binary object-existence task. The model is asked whether a specific object appears in the image, and the answer is normalized as either “yes” or “no”. This design makes POPE suitable for measuring whether a model incorrectly confirms the presence of objects that are not actually visible in the image [3].

Table 4.2: Key characteristics of the POPE dataset used in this thesis.

Characteristic	Description
Main purpose	Evaluation of object hallucination in image-based VLMs
Question format	Binary yes/no object-existence questions
Positive questions	Ask about objects that are present in the image
Negative questions	Ask about objects that are not present in the image
Metrics used	Accuracy, precision, recall, F1, yes ratio, and confusion counts

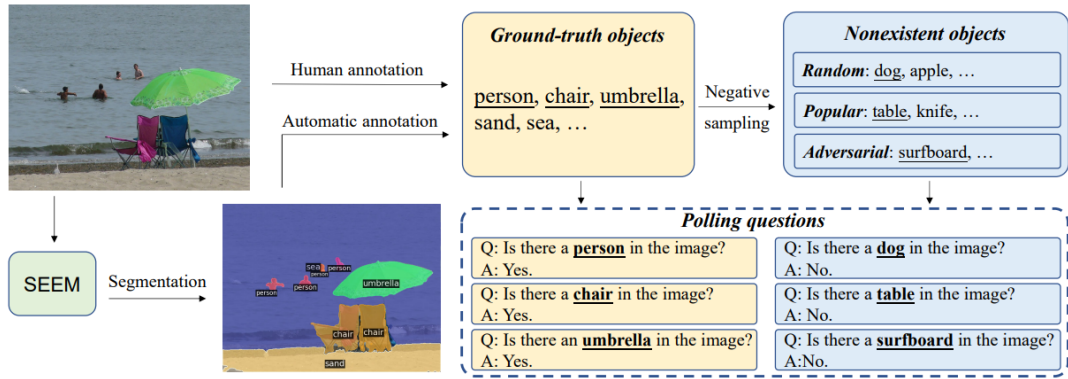


Figure 4.2: Illustration of the POPE object-existence probing process.

Figure 4.2 illustrates how POPE constructs object-existence questions. Ground-truth objects are first identified from the image using human or automatic annotations. Positive questions are generated from objects that appear in the image, while negative questions are generated from nonexistent objects sampled under random, popular, or adversarial settings. This design is relevant to this thesis because visual-token pruning may affect whether the model retains enough evidence to reject nonexistent objects instead of hallucinating their presence.

4.1.3 Failure-Mining Set

In addition to GQA and POPE, this thesis uses a fixed failure-mining set for diagnostic analysis. This set contains 100 cases stored in a CSV manifest. The cases are drawn from GQA validation, VQAv2 validation, and TextVQA validation sources. The largest portion comes from GQA validation, while VQAv2 and TextVQA cases are included to provide additional diagnostic coverage.

The failure-mining set is not intended to represent the full distribution of any benchmark dataset. Instead, it is used as a targeted stress-test set for analyzing failure patterns under sparse visual-token selection. The set contains cases related to visual evidence such as object existence, attributes, relations, counting, and OCR-related information.

After inference, failure-driven analysis focuses on cases where SparseVLM-Original under an aggressive sparse setting gives an incorrect answer. These cases are then used to examine whether alternative selection methods recover the correct answer and whether the selected visual tokens provide more useful visual evidence. Therefore, the failure-mining set is used for diagnostic recovery analysis and qualitative visualization, not as a replacement for standard benchmark evaluation.

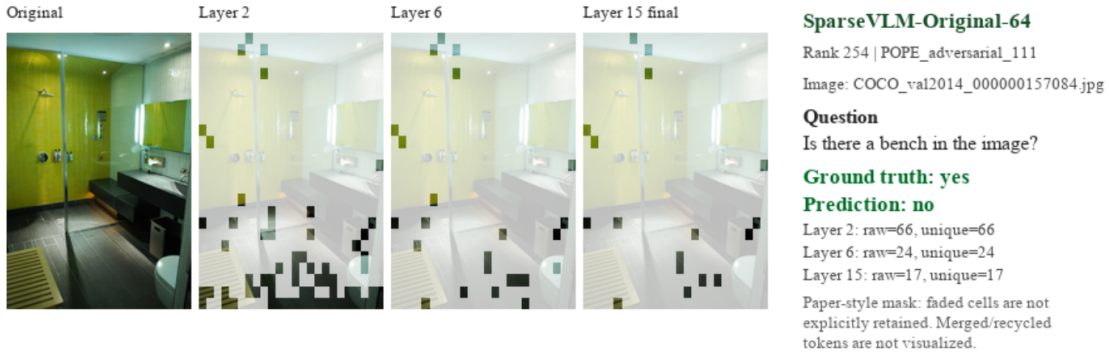


Figure 4.3: Example visualization of an image from the failure-mining set.

4.2 Methods Compared

This thesis compares five inference settings. The goal is not only to compare answer-level performance, but also to understand how different visual-token selection strategies behave under sparse inference. The compared methods are summarized in Table 4.3.

Dense / Vanilla is used as the full-token reference, while SparseVLM-Original provides the main sparse baseline. Ours and Threshold-Fixed are compared under the same nominal sparse-budget settings to evaluate fixed-budget redundancy-aware selection. Threshold-Adaptive is not included in the main fixed-budget comparison because it does not enforce the same target selected-token budget as SparseVLM-Original, Ours, or Threshold-Fixed. Instead, it is evaluated separately as a variable-token baseline to study how a simple similarity-threshold rule trades retained-token count for answer-level performance.

Detailed descriptions of the selection mechanisms are provided in Chapter 3. The remaining sections of this chapter describe the token-budget settings, evaluation metrics, and evaluation protocol used to compare these methods.

Table 4.3: Methods compared in the experiments.

Method	Role in the experiments
Dense / Vanilla	Full-token reference setting. It keeps the complete visual-token sequence and serves as the upper reference for sparse methods.
SparseVLM-Original	Main sparse baseline. It selects visual tokens according to text-guided relevance scores.
Ours	Proposed redundancy-aware method. It keeps the SparseVLM-style text-guided relevance signal but adds a redundancy-aware selection mechanism.
Threshold-Fixed	Fixed-budget threshold-based baseline. It uses hard redundancy filtering while preserving the same target selected-token budget as the fixed-budget sparse methods.
Threshold-Adaptive	Variable-token threshold-based baseline. It allows the number of retained visual tokens to vary according to the similarity threshold and is used to study the token–performance trade-off.

4.3 Token Budgets

The experiments distinguish between the full-token reference setting and sparse settings controlled by nominal retained-token configurations. In the implementation, these configurations are stored as `retained_tokens` in run manifests and reported as `token_setting` in the summary files. These values should be interpreted as sparse-budget configuration labels, rather than exact final visual-token sequence lengths.

4.3.1 Full-Token Setting

The Dense / Vanilla baseline uses the full visual-token sequence produced by the vision encoder. In the evaluated configuration, this corresponds to 576 visual tokens. This setting does not apply sparse visual-token selection and is used as the full-token reference point for comparison with sparse methods.

4.3.2 Fixed Sparse-Budget Settings

For the fixed-budget sparse methods, the main experiments use two nominal sparse-budget settings: 128 and 64. These settings apply to the main fixed-budget sparse methods: SparseVLM-Original, the proposed method, and Threshold-Fixed. They represent target sparse configurations rather than exact final token counts.

The implementation maps each nominal sparse setting to a layer-wise keep schedule over the sparse pruning layers. The nominal 128 setting maps to the keep schedule [303, 110, 36], while the nominal 64 setting maps to [66, 30, 17]. These values correspond to the sparse layers used by the SparseVLM pipeline. As a result,

the nominal 64 and 128 settings should not be interpreted as exact final visual-token sequence lengths or as exact counts of original image-patch tokens.

Table 4.4: Token-budget configurations used in the experiments. Sparse-budget values are nominal settings and may differ from the final visual-token sequence length after token recycling/compression.

Setting	Reported setting	token	Interpretation
Dense / Vanilla	576		Full visual-token sequence without sparse selection
Nominal sparse budget	128		Fixed sparse setting with layer-wise keep schedule [303, 110, 36]
Nominal sparse budget	64		Fixed sparse setting with layer-wise keep schedule [66, 30, 17]
Threshold-Adaptive	Variable-token setting		Threshold-controlled selected-token count without fixed-budget enforcement

4.3.3 Variable-Token Setting

Threshold-Adaptive is treated separately from the fixed-budget sparse methods. Although its runs are associated with a nominal anchor setting in the implementation, the method does not enforce the same fixed selected-token budget as SparseVLM-Original, Ours, or Threshold-Fixed. Instead, it filters relevance-ordered candidates using a similarity threshold, so the number of selected current visual tokens may vary across examples and layers.

After the adaptive threshold mask is produced, the method still follows the same SparseVLM sparse inference pipeline, including token recycling and compression where applicable. Therefore, Threshold-Adaptive should be interpreted as a variable-token threshold baseline for trade-off analysis rather than as a fixed-budget method.

4.4 Evaluation Metrics

This thesis uses five groups of evaluation metrics: benchmark task metrics, inference latency, failure-driven diagnostic metrics, auxiliary token-selection metrics, and qualitative visualization criteria. Benchmark task metrics measure answer-level performance on GQA and POPE. Inference latency measures the efficiency side of sparse visual-token selection in the local experimental environment. Failure-driven metrics examine whether alternative sparse selection methods recover cases where SparseVLM-Original fails. Auxiliary metrics describe the behavior of the selected visual tokens. Visualization criteria provide qualitative evidence for interpreting token-selection behavior.

4.4.1 Benchmark Task Metrics

GQA. Performance is measured using normalized exact-match accuracy. A prediction is counted as correct when the normalized predicted answer matches the normalized ground-truth answer. The answer-normalization process is applied as follows:

- convert the predicted and ground-truth answers to lowercase;
- remove leading and trailing whitespace;
- remove a trailing period when present;
- compare the normalized prediction with the normalized ground-truth answer.

The resulting metric is reported as **GQA accuracy**. This metric is appropriate because GQA answers are usually short, such as object names, attributes, numbers, or binary responses.

POPE. Evaluation is treated as binary classification over yes/no object-existence questions. The prediction-normalization process is applied as follows:

- if the normalized model response contains a negative expression such as “no” or “not”, it is mapped to `no`;
- otherwise, the response is mapped to `yes`;
- the normalized binary prediction is compared with the binary ground-truth label.

The main POPE metrics are **accuracy** and **F1 score**. Precision, recall, yes ratio, and confusion counts are also reported as supporting metrics, because they help identify whether a method tends to over-answer “yes” or miss objects that are actually present.

4.4.2 Inference Latency Metric

In addition to answer-level performance, this thesis reports inference latency to evaluate the efficiency side of sparse visual-token selection. Latency is measured as the mean inference time per sample, reported in seconds per sample:

$$T_{\text{avg}} = \frac{1}{N} \sum_{i=1}^N T_i,$$

where T_i is the inference time for sample i , and N is the number of evaluated samples.

This metric is used to compare the computational behavior of Dense / Vanilla, SparseVLM-Original, Ours, and Threshold-Fixed on the main GQA and POPE evaluation settings. Since the measurements are collected in the local experimental environment, they should be interpreted as relative latency results under the same hardware and implementation setup, not as hardware-independent runtime benchmarks.

4.4.3 Failure-Driven Diagnostic Metrics

The failure-driven analysis focuses on cases where SparseVLM-Original gives an incorrect answer under an aggressive sparse setting. These cases are used to examine whether alternative selection strategies can recover failures caused by sparse visual-token selection.

The main diagnostic metrics are:

- **Recovery:** a case where SparseVLM-Original is incorrect, but the compared method gives the correct answer.
- **Regression:** a case where SparseVLM-Original is correct, but the compared method gives an incorrect answer.

Based on these definitions, the analysis reports the number of recovered failures, the recovery rate, the number of regressions, and the net change between recoveries and regressions. These metrics are used only for diagnostic interpretation and should not be treated as a replacement for benchmark-level evaluation.

4.4.4 Auxiliary Token-Selection Metrics

In addition to answer-level performance, this thesis computes several auxiliary metrics to better understand the behavior of sparse visual-token selection. These metrics are not direct measures of task accuracy. Instead, they provide diagnostic information about the selected token sets.

- **Pairwise hidden-state similarity:** measures the average cosine similarity among selected visual-token representations. It is used as a proxy for redundancy, where lower similarity suggests that selected tokens may contain less overlapping information. Given a selected token set $\mathcal{S}$, the average pairwise similarity is computed as:

$$\text{Sim}_{\text{avg}}(\mathcal{S}) = \frac{2}{|\mathcal{S}|(|\mathcal{S}| - 1)} \sum_{i < j, v_i, v_j \in \mathcal{S}} \text{sim}(v_i, v_j),$$

where $\text{sim}(v_i, v_j)$ denotes cosine similarity between the hidden-state representations of two selected visual tokens.

- **Spatial coverage:** measures how selected original patch identifiers are distributed over the 24×24 visual grid. Let $\mathcal{P}$ be the set of selected original patches, where each patch $p \in \mathcal{P}$ has a grid coordinate (x_p, y_p) . The analysis uses indicators such as unique patch count, grid occupancy, bounding-box area, and centroid distance:

$$U = |\mathcal{P}|, \quad \text{Occ}(\mathcal{P}) = \frac{|\mathcal{P}|}{24 \times 24},$$

$$A_{\text{bbox}} = \frac{(x_{\max} - x_{\min} + 1)(y_{\max} - y_{\min} + 1)}{24 \times 24},$$

$$d_{\text{centroid}} = \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} \|(x_p, y_p) - \bar{p}\|_2, \quad \bar{p} = \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} (x_p, y_p).$$

These quantities describe whether selected patches are spatially concentrated or spread across the image.

- **Selection overlap analysis:** compares how much the selected original patch sets coincide across different methods. For two methods A and B , let $\mathcal{P}_A$ and $\mathcal{P}_B$ denote their selected patch sets. The raw overlap and Jaccard similarity are defined as:

$$O(A, B) = |\mathcal{P}_A \cap \mathcal{P}_B|,$$

$$J(A, B) = \frac{|\mathcal{P}_A \cap \mathcal{P}_B|}{|\mathcal{P}_A \cup \mathcal{P}_B|}.$$

In this thesis, Jaccard similarity is used as the normalized overlap measure because it makes method pairs easier to compare across datasets and sparse-budget settings.

These metrics must be interpreted carefully. For example, lower pairwise similarity may indicate reduced redundancy, but it does not automatically imply better answer-level performance. Similarly, broader spatial coverage or lower selection overlap may indicate different token-selection behavior, but these signals do not directly prove better visual reasoning.

4.4.5 Qualitative Visualization Criteria

Visualization is used to inspect selected original visual patches for representative cases. The selected patch identifiers are mapped onto a 24×24 image grid to show which regions are retained by each method. These visualizations are evaluated qualitatively using the following criteria:

- whether the selected tokens cover the hypothesized answer-relevant evidence region;

- whether the selected tokens appear concentrated in a potentially redundant region;
- whether the visualization supports the interpretation of a recovered or failed case;
- whether the evidence remains unclear from the visualization alone.

The visualization analysis is used only as qualitative support. It can illustrate plausible token-selection behavior, but it does not prove that a specific selected patch caused an answer change. Merged or recycled tokens are also not visualized as original patches, because they do not correspond to a single original image location.

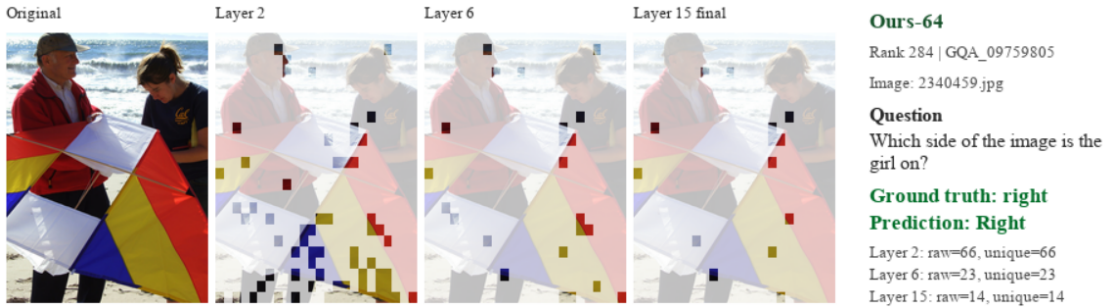


Figure 4.4: Example visualization of an image from the failure-mining set.

Figure 4.4 shows how retained original visual patches are visualized across sparse layers together with the corresponding question, ground-truth answer, model prediction, and selected-token counts.

4.5 Evaluation Protocol

The evaluation protocol is organized into five complementary parts. First, the main benchmark evaluation measures answer-level performance on fixed GQA and POPE evaluation subsets. Second, Threshold-Adaptive is evaluated separately as a variable-token trade-off baseline. Third, the failure-driven analysis examines targeted cases where SparseVLM-Original fails under an aggressive sparse setting. Fourth, the visualization protocol provides qualitative inspection of selected visual tokens. Finally, the ablation and auxiliary-metric analysis studies how different redundancy-aware settings affect selection behavior. This structure separates benchmark comparison, variable-token trade-off analysis, targeted error analysis, qualitative inspection, and mechanism-level diagnostics.

4.5.1 Main Benchmark Evaluation

The main benchmark evaluation measures answer-level performance on fixed local evaluation subsets of GQA and POPE. Its purpose is to compare the full-token

reference setting with sparse visual-token selection methods under controlled and reproducible conditions. The protocol is summarized in Table 4.5.

Table 4.5: Summary of the main benchmark evaluation protocol.

Benchmark	Compared settings	Main metrics
GQA	Dense / Vanilla	Accuracy
	SparseVLM-Original	
	Ours	
	Threshold-Fixed	
POPE	Dense / Vanilla	Accuracy and F1
	SparseVLM-Original	
	Ours	
	Threshold-Fixed	

In addition to answer-level metrics, the main benchmark evaluation also records per-sample inference latency for GQA and POPE. Latency is reported for Dense / Vanilla, SparseVLM-Original, Ours, and Threshold-Fixed under the same local experimental environment. These results are used to analyze the accuracy–efficiency trade-off introduced by sparse visual-token selection.

4.5.2 Threshold-Adaptive Trade-off Evaluation

Threshold-Adaptive is evaluated separately from the main fixed-budget comparison. Unlike Threshold-Fixed, it does not use backfill to match a predetermined selected-token budget. Instead, the number of retained visual tokens depends on the similarity threshold τ . Therefore, its results are interpreted as a variable-token trade-off rather than as a directly budget-matched comparison with SparseVLM-Original, Ours, or Threshold-Fixed.

The Threshold-Adaptive evaluation reports answer-level performance together with the average retained-token count. In the main analysis, Threshold-Adaptive is compared against Ours-64 and Ours-128 as reference points, because its average retained-token count falls between these two fixed-budget settings. The evaluated threshold values are $\tau = 0.80$, $\tau = 0.85$, and $\tau = 0.90$.

For GQA, the main metric is accuracy. For POPE, the main metrics are accuracy and F1. On the failure-mining set, Threshold-Adaptive is used as an additional diagnostic comparison to examine whether variable-token threshold filtering can recover SparseVLM-Original failures under aggressive sparse inference.

4.5.3 Failure-Driven Analysis

The failure-driven analysis uses the curated failure-mining set described in Section 4.1.3. Unlike the main benchmark evaluation, this analysis is designed as a diagnostic stress test. Its purpose is to examine whether alternative sparse selection

strategies can recover cases where SparseVLM-Original fails under an aggressive nominal sparse-budget setting.

The experimental arrangement is summarized in Table 4.6.

Table 4.6: Summary of the failure-driven analysis protocol.

Component	Protocol
Evaluation set	Curated failure-mining set used as a diagnostic stress test
Compared methods	Dense / Vanilla, SparseVLM-Original, Ours, and Threshold-Fixed
Main reported metrics	Accuracy, recovered failures, recovery rate, regressions, and net recovery-regression change

For each case, SparseVLM-Original is treated as the main sparse reference method, and its incorrect predictions define the baseline failure cases for recovery analysis. Ours and Threshold-Fixed are then evaluated according to the diagnostic metrics defined in Section 4.4.

4.5.4 Visualization Protocol

The visualization protocol inspects selected original visual patches for representative cases. The visualization scripts map selected original CLIP patch indices onto a 24-by-24 image grid and render method comparisons for SparseVLM-Original, Ours, and Threshold-Fixed. The final SparseVLM-style figures show layer-wise retained original patches at the sparse pruning layers.

Merged or recycled tokens are not visualized because they do not correspond to a single original image patch.

4.5.5 Ablation and Auxiliary Metrics

The ablation analysis studies how the proposed redundancy-aware selection method behaves under different hyperparameter settings. It uses deterministic 500-sample subsets of GQA and POPE generated with the same fixed seed. These subsets are used only for ablation and diagnostic analysis, not as replacements for the main benchmark evaluation.

The ablation focuses on two hyperparameters:

- **candidate_pool_factor:** controls the size of the relevance-based candidate pool before redundancy-aware selection is applied. A larger value gives the method more candidate tokens to choose from, which may increase diversity but can also introduce less relevant candidates.
- **lambda_relevance:** controls the balance between text-guided relevance

and redundancy reduction in Ours. A larger value places more emphasis on relevance, while a smaller value gives more influence to the redundancy penalty.

The ablation compares different Ours variants against SparseVLM-Original and Threshold-Fixed on the same deterministic subsets. This setup makes it possible to examine whether changes in the redundancy-aware selection mechanism lead to different answer-level performance and token-selection behavior.

In addition to answer-level metrics, the auxiliary analysis reports the following diagnostic metrics:

- **Pairwise hidden-state similarity:** used as a proxy for redundancy among selected visual tokens.
- **Spatial coverage:** used to inspect how selected original patches are distributed over the visual grid.
- **Overlap and Jaccard similarity:** used to compare how much the selected patch sets overlap across methods.

These auxiliary metrics are used to support interpretation of the selection mechanism. However, they are not treated as independent evidence of better task performance. In particular, lower pairwise similarity or broader spatial coverage may indicate different token-selection behavior, but they do not automatically imply higher accuracy.

CHAPTER 5. Results and Analysis

5.1 Main Quantitative Results

This section presents the main quantitative results of the evaluated methods. The analysis focuses on three evaluation settings: GQA, POPE, and the failure-mining diagnostic set. GQA and POPE are used to evaluate general benchmark performance, while the failure-mining set is used to examine targeted sparse-selection failures. The fixed-budget comparison includes Dense / Vanilla, the original SparseVLM baseline, Ours, and Threshold-Fixed. Threshold-Adaptive is not included in the same fixed-budget tables because it is a variable-token method and is analyzed separately as a token–performance trade-off baseline.

5.1.1 GQA Results

Table 5.1 reports the GQA results. The Dense / Vanilla baseline achieves the highest overall accuracy, as expected, because it uses the full visual-token sequence. Among sparse methods, reducing the nominal sparse-budget setting leads to a clear performance drop, especially under the aggressive sparse setting.

Table 5.1: GQA accuracy results. Accuracy is reported as normalized exact-match accuracy. The improvement column compares each method against SparseVLM-Original under the same nominal sparse-budget setting.

Method	Budget setting	Accuracy (%)	Improvement
Dense / Vanilla	576	71.48	–
SparseVLM-Original	128	66.08	–
Ours	128	67.62	+1.54 pp
Threshold-Fixed	128	68.10	+2.02 pp
SparseVLM-Original	64	60.62	–
Ours	64	62.48	+1.86 pp
Threshold-Fixed	64	63.20	+2.58 pp

All sparse methods reduce GQA accuracy compared with the full-token reference. Dense / Vanilla achieves the highest GQA accuracy with 71.48%, while all sparse methods obtain lower accuracy. This shows that reducing the number of visual tokens introduces an information-preservation challenge. Although sparse selection improves efficiency, removing visual tokens can also remove evidence needed for answering GQA questions correctly.

Ours improves GQA accuracy at both sparse-budget settings. Compared with SparseVLM-Original, Ours improves GQA accuracy by +1.54 percentage points at the nominal 128-token setting and by +1.86 percentage points at the nominal 64-token setting. This indicates that adding redundancy awareness to the

SparseVLM-style selection rule can improve over direct relevance-based TopK selection on GQA.

Threshold-Fixed achieves the strongest sparse GQA accuracy in this comparison. At the nominal 128-token setting, Threshold-Fixed reaches 68.10%, which is higher than both SparseVLM-Original and Ours. At the nominal 64-token setting, it also achieves the best sparse accuracy with 63.20%. Therefore, the GQA results support the usefulness of redundancy-aware selection, but they do not show that Ours is the best sparse method overall.

5.1.2 POPE Results

Table 5.2 reports the POPE results. Since POPE uses yes/no questions about object presence, both accuracy and F1 are reported. The Dense / Vanilla baseline again provides the strongest full-token reference point.

Table 5.2: POPE results. Accuracy and F1 are reported for the yes/no object-hallucination evaluation.

Method	Budget setting	Accuracy (%)	F1 (%)
Dense / Vanilla	576	86.53	87.61
SparseVLM-Original	128	85.33	86.32
Ours	128	85.27	86.60
Threshold-Fixed	128	85.07	86.41
SparseVLM-Original	64	81.00	81.81
Ours	64	83.00	84.09
Threshold-Fixed	64	84.33	85.34

At the nominal 128-token setting, the effect of Ours on POPE is mixed. Compared with SparseVLM-Original, Ours slightly decreases accuracy from 85.33% to 85.27%, but improves F1 from 86.32% to 86.60%. This suggests that, under the less aggressive sparse setting, the redundancy-aware selection rule does not clearly improve all POPE metrics, even though it can improve the balance between precision and recall reflected by F1.

Ours provides a clearer improvement under the aggressive nominal 64-token setting. Compared with SparseVLM-Original, Ours improves POPE accuracy from 81.00% to 83.00% and F1 from 81.81% to 84.09%. This corresponds to gains of +2.00 percentage points in accuracy and +2.28 percentage points in F1. The improvement is especially meaningful because POPE evaluates object-existence hallucination, where losing relevant visual evidence may cause the model to incorrectly answer yes or no.

Threshold-Fixed remains a strong threshold-based baseline on POPE. At the nominal 64-token setting, Threshold-Fixed achieves the strongest sparse POPE

performance with 84.33% accuracy and 85.34% F1. At the nominal 128-token setting, no single sparse method dominates all metrics: SparseVLM-Original has the highest sparse accuracy, while Ours has the highest sparse F1. Overall, POPE suggests that redundancy-aware selection is more useful under stronger pruning, but the best redundancy-aware variant depends on the evaluation setting.

5.1.3 Failure-Mining Results

Table 5.3 summarizes the results on the failure-mining diagnostic set. Unlike GQA and POPE, this set is not intended to represent a full benchmark distribution. It is used to stress-test sparse visual-token selection and to identify whether alternative selection methods can recover cases where SparseVLM-Original fails.

Table 5.3: Failure-mining diagnostic results. Recovery and regression counts are computed relative to SparseVLM-Original under the same nominal sparse-budget setting.

Method	Budget setting	Correctness (%)	Recovered	Regressions
Dense / Vanilla	576	78.00	–	–
SparseVLM-Original	128	79.00	–	–
Ours	128	79.00	2	2
Threshold-Fixed	128	74.00	2	7
SparseVLM-Original	64	70.00	–	–
Ours	64	75.00	10	5
Threshold-Fixed	64	70.00	7	7

Ours gives the clearest improvement under the aggressive nominal 64-token setting. At the nominal 64-token setting, SparseVLM-Original achieves 70.00% accuracy, while Ours improves the result to 75.00%. In terms of diagnostic recovery behavior, Ours recovers 10 SparseVLM-Original failures and introduces 5 regressions, giving a positive net recovery-regression change. This suggests that the redundancy-aware selection rule can help recover some cases where direct relevance-based selection fails under aggressive pruning.

Threshold-Fixed does not show the same gain on the failure-mining set. At the nominal 64-token setting, Threshold-Fixed achieves 70.00% accuracy, which is the same as SparseVLM-Original on this diagnostic set. It recovers 7 SparseVLM-Original failures, but also introduces 7 regressions. This indicates that hard-threshold filtering can recover some failure cases, but its benefits are offset by newly introduced errors in this setting.

The nominal 128-token setting shows limited separation among methods. At the nominal 128-token setting, SparseVLM-Original and Ours both achieve 79.00% accuracy, while Threshold-Fixed decreases to 74.00%. This suggests that the benefit of Ours is more visible when pruning is more aggressive. When more

visual tokens are retained, the redundancy-aware modification may have less room to improve over the SparseVLM-Original selection behavior.

Overall, the failure-mining results support the usefulness of Ours for targeted failure recovery. The strongest evidence from this diagnostic set is not that Ours is universally better, but that it can recover a meaningful subset of SparseVLM-Original failures under aggressive sparse inference. This complements the GQA and POPE results by showing that redundancy-aware selection may be especially useful for difficult cases where sparse token selection removes or underrepresents relevant visual evidence.

5.1.4 Variable-Token Threshold-Adaptive Analysis

Threshold-Adaptive is analyzed separately from the fixed-budget sparse methods because it does not enforce a fixed selected-token budget. Instead, it applies a similarity threshold and allows the number of retained tokens to vary across examples. Therefore, the results in this subsection should be interpreted as a token–performance trade-off, not as a directly budget-matched comparison with Ours, SparseVLM-Original, or Threshold-Fixed.

Table 5.4 summarizes the Threshold-Adaptive results on GQA and POPE, together with Ours-64 and Ours-128 as fixed-budget reference points.

Table 5.4: Threshold-Adaptive trade-off results compared with Ours. Threshold-Adaptive uses a variable retained-token count and is therefore not a fixed-budget baseline.

Dataset	Method	τ	Accuracy	F1	Avg. retained tokens
GQA	Ours-64	–	0.6248	–	18.00
GQA	Ours-128	–	0.6762	–	39.00
GQA	Threshold-Adaptive	0.80	0.6284	–	33.18
GQA	Threshold-Adaptive	0.85	0.6368	–	33.68
GQA	Threshold-Adaptive	0.90	0.6406	–	34.10
POPE	Ours-64	–	0.8300	0.8409	18.00
POPE	Ours-128	–	0.8527	0.8660	39.00
POPE	Threshold-Adaptive	0.80	0.8240	0.8335	33.18
POPE	Threshold-Adaptive	0.85	0.8340	0.8415	33.62
POPE	Threshold-Adaptive	0.90	0.8353	0.8432	34.11

Threshold-Adaptive provides an intermediate variable-token setting. The recorded average retained-token count of Threshold-Adaptive is around 33–34 tokens. This is higher than Ours-64, which retains 18 tokens on average in the reported setting, but lower than Ours-128, which retains 39 tokens on average. Therefore, Threshold-Adaptive should not be read as a direct competitor to either Ours-64 or Ours-128. Instead, it provides an intermediate trade-off point between the two fixed-budget settings.

On GQA, increasing the threshold improves accuracy while retaining slightly more tokens. When τ increases from 0.80 to 0.90, the average retained-token count increases from 33.18 to 34.10, and GQA accuracy improves from 0.6284 to 0.6406. This indicates that a more permissive threshold allows the method to preserve more visual information and recover some performance. However, even at $\tau = 0.90$, Threshold-Adaptive remains below Ours-128, which achieves 0.6762 accuracy.

On POPE, higher-threshold Threshold-Adaptive slightly exceeds Ours-64 but remains below Ours-128. At $\tau = 0.85$ and $\tau = 0.90$, Threshold-Adaptive achieves higher accuracy and F1 than Ours-64. However, these results are obtained with a higher average retained-token count than Ours-64. Threshold-Adaptive also remains below Ours-128 in both accuracy and F1. This suggests that variable-token threshold filtering can recover part of the performance lost under aggressive sparsification, but it does so by retaining more tokens.

5.1.5 Inference Latency Results

Besides answer-level performance, latency is measured to evaluate the efficiency side of sparse visual-token selection. Figure 5.1 shows the mean per-sample inference latency of Dense / Vanilla, SparseVLM-Original, Ours, and Threshold-Fixed on GQA and POPE. These latency values are measured in the local experimental environment and should be interpreted as relative runtime results under the same hardware and implementation setup.

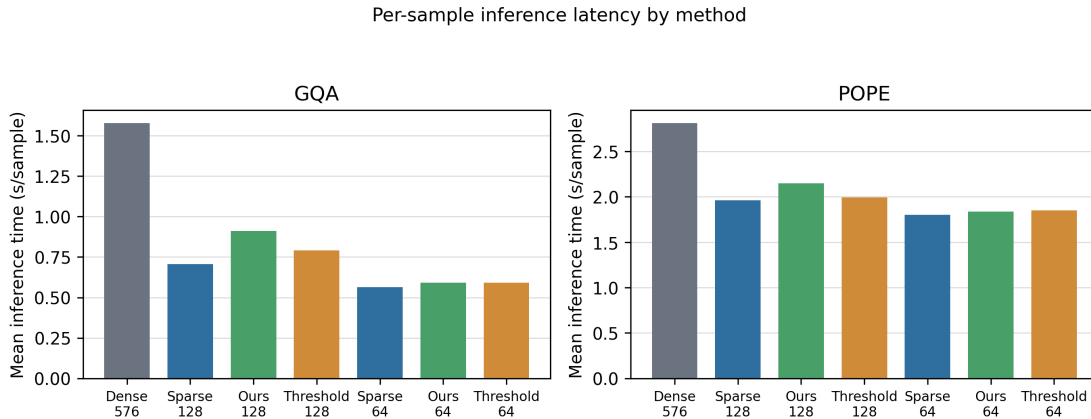


Figure 5.1: Per-sample inference latency by method on GQA and POPE.

Sparse inference substantially reduces latency compared with Dense / Vanilla. On GQA, Dense / Vanilla requires 1.579 seconds per sample, while SparseVLM-Original reduces latency to 0.707 seconds per sample at the nominal 128-token setting and 0.563 seconds per sample at the nominal 64-token setting. On POPE, Dense / Vanilla requires 2.813 seconds per sample, while SparseVLM-Original reduces latency to 1.961 seconds per sample at the nominal 128-token setting and

1.803 seconds per sample at the nominal 64-token setting. These results confirm that sparse visual-token selection provides a clear efficiency benefit compared with full-token inference.

Ours introduces additional selection overhead compared with SparseVLM-Original. At the nominal 128-token setting, Ours takes 0.911 seconds per sample on GQA, compared with 0.707 seconds for SparseVLM-Original. On POPE, Ours takes 2.148 seconds per sample, compared with 1.961 seconds for SparseVLM-Original. This overhead is expected because Ours performs redundancy-aware selection rather than direct relevance-based TopK selection.

The overhead of Ours is smaller under the aggressive nominal 64-token setting. At the nominal 64-token setting, Ours requires 0.592 seconds per sample on GQA, compared with 0.563 seconds for SparseVLM-Original. On POPE, Ours requires 1.836 seconds per sample, compared with 1.803 seconds for SparseVLM-Original. This suggests that the additional cost of redundancy-aware selection becomes more modest when the selected-token budget is smaller.

Threshold-Fixed also remains faster than Dense / Vanilla while adding limited overhead. Threshold-Fixed requires 0.791 seconds per sample on GQA and 1.995 seconds per sample on POPE at the nominal 128-token setting. At the nominal 64-token setting, it requires 0.592 seconds per sample on GQA and 1.850 seconds per sample on POPE. These values show that threshold-based redundancy filtering also preserves much of the latency advantage of sparse inference.

Overall, latency results should be interpreted together with answer-level performance. SparseVLM-Original is generally the fastest sparse method, while Ours and Threshold-Fixed introduce some additional overhead due to redundancy-aware selection. However, Ours still remains substantially faster than Dense / Vanilla and improves answer-level performance over SparseVLM-Original in several sparse settings. Therefore, Ours should be interpreted as an accuracy–efficiency trade-off method rather than as the fastest sparse inference method.

5.1.6 Summary of Main Results

The main quantitative results can be summarized into six findings:

1. **Dense / Vanilla remains the strongest full-token reference.** Across the main benchmark settings, the full-token setting generally achieves the highest performance. This confirms that sparse visual-token selection introduces an information-preservation challenge.
2. **Ours improves over SparseVLM-Original in several sparse settings.** The

proposed redundancy-aware method improves GQA accuracy at both nominal sparse-budget settings and gives a clearer improvement on POPE under the aggressive nominal 64-token setting.

3. **Threshold-Fixed is a strong fixed-budget redundancy-aware baseline.** Threshold-Fixed achieves the strongest sparse performance in several GQA and POPE settings. Therefore, the results support redundancy-aware selection in general, but they do not show that Ours is universally superior to all alternatives.
4. **Sparse methods substantially reduce inference latency compared with Dense / Vanilla.** SparseVLM-Original is generally the fastest sparse method, while Ours and Threshold-Fixed introduce additional selection overhead. However, both redundancy-aware methods still remain substantially faster than Dense / Vanilla, so their results should be interpreted in terms of an accuracy–efficiency trade-off.
5. **Threshold-Adaptive provides a useful variable-token trade-off baseline.** Threshold-Adaptive retains more tokens than Ours-64 but fewer than Ours-128 on average. Its results show that a simple similarity-threshold rule can recover part of the performance lost under aggressive sparsification, but it is not directly comparable to fixed-budget methods and does not replace the main fixed-budget comparison.
6. **The failure-mining results highlight the diagnostic value of Ours.** On the curated failure-mining set, Ours gives the clearest improvement under the aggressive nominal 64-token setting and recovers more SparseVLM-Original failures than Threshold-Fixed. This suggests that Ours may be especially useful for targeted failure recovery under aggressive sparse inference.

Therefore, the main results support a careful claim: the proposed redundancy-aware selection method improves over SparseVLM-Original, especially under aggressive sparse pruning and targeted failure cases. However, the results do not support the stronger claim that Ours is universally better than all redundancy-aware baselines. The following sections analyze these findings in more detail through failure-driven quantitative and qualitative analysis, ablation, and auxiliary token-selection metrics.

5.2 Failure-Driven Analysis

The main quantitative results show that the proposed method is most useful under the aggressive nominal 64-token setting, especially on the failure-mining diagnostic set. This section analyzes those failures in more detail. The goal is not

to establish a new benchmark result, but to understand what kinds of SparseVLM-Original failures can be recovered by alternative token-selection strategies.

The analysis focuses on cases where SparseVLM-Original under the nominal 64-token setting gives an incorrect answer. This produces a set of 30 baseline failure cases. These cases are then compared against Ours and Threshold-Fixed under the same nominal sparse setting. A case is counted as recovered if SparseVLM-Original is incorrect while the comparison method is correct. A case is counted as a regression if SparseVLM-Original is correct while the comparison method is incorrect.

5.2.1 Quantitative Results

Table 5.5: Failure recovery comparison under the aggressive nominal 64-token setting. Recovery is computed over SparseVLM-Original failure cases.

Method	Recovered ures	fail- Regressions	Net gain
Ours	10	5	+5
Threshold-Fixed	7	7	0

Table 5.5 shows that Ours recovers more SparseVLM-Original failures than Threshold-Fixed while introducing fewer regressions. Ours recovers 10 of the 30 SparseVLM-Original failure cases, corresponding to a recovery rate of 33.3%. Threshold-Fixed recovers 7 cases but also introduces 7 regressions, resulting in no net gain over SparseVLM-Original on this diagnostic set. In contrast, Ours produces a positive net gain of 5 cases.

The overlap between recovered cases also provides useful information. Among the 30 SparseVLM-Original failures, both methods recover 7 cases. Ours alone recovers 3 additional cases, while Threshold-Fixed alone recovers none. The remaining 20 cases are not fixed by either redundancy-aware method.

Table 5.6: Overlap of recovered SparseVLM-Original failures under the nominal 64-token setting.

Recovery category	Number of cases
Recovered by both Ours and Threshold-Fixed	7
Recovered only by Ours	3
Recovered only by Threshold-Fixed	0
Not recovered by either method	20

This overlap pattern suggests that the proposed method is not merely duplicating the behavior of Threshold-Fixed. Both methods can recover some of the same failures, which is expected because both introduce redundancy-aware selection compared with direct TopK selection. However, the presence of Ours-only recoveries

indicates that the continuous relevance–redundancy trade-off can select a different useful subset of visual tokens from the hard-threshold baseline.

5.2.2 Qualitative Results

To better understand the recovered cases, a manual review was conducted on representative visualizations from the failure-driven analysis. The review focused on whether the selected original visual patches appeared to cover the visual evidence needed to answer the question.

In the manually reviewed failure-mining recovery cases, only one case showed clear visual support for the interpretation that Ours preserved more answer-relevant evidence than SparseVLM-Original. The remaining six reviewed cases were unclear. In these unclear cases, the visualization did not provide strong evidence that Ours focused more directly on the relevant region, even when the answer-level result improved.

The manual review was then expanded to additional recovery candidates from GQA and POPE. The same pattern remained: only a small number of recovered examples showed clear visual support, while most cases remained visually ambiguous. In many examples, the selected-token overlays did not clearly indicate that Ours concentrated more strongly on the object or region required by the question.

5.2.3 Interpretation and Limitations

This result does not mean that the recovered answers are unimportant. Rather, it shows that patch-level visualization provides only limited evidence about why an answer changed. A method may recover an answer even when the selected patch overlay does not visibly highlight the expected region in a clear and human-interpretable way.

There are several reasons for this limitation. First, selected patch overlays show which original patches are retained, but they do not reveal how the language model internally uses those visual representations. A retained patch may influence the answer indirectly, and an apparently relevant patch may not be used strongly by the model.

Second, the visualization only shows selected original patches. Recycled or compressed tokens are not visualized as single image patches because they do not correspond to one original patch location. Therefore, the displayed overlay is only a partial view of the information available to the model after sparse selection and token recycling.

Third, some answer changes may result from small differences in selected evi-

dence, language-model sensitivity, or interactions with token recycling. These effects may not be visible from patch overlays alone. As a result, qualitative visualization cannot establish a direct causal link between a selected patch and a corrected answer.

The main conclusion from the qualitative analysis is therefore cautious. The failure-driven results show that Ours can recover some SparseVLM-Original failures under aggressive sparse inference. However, the visualizations provide clear support for this interpretation only in a limited number of cases. Most reviewed cases remain unclear, so the visualization analysis should be treated as diagnostic support rather than proof of the mechanism. This reinforces the need to interpret the failure-driven results together with quantitative recovery metrics and auxiliary token-selection metrics.

5.3 Ablation Study

This section analyzes how two main hyperparameters affect answer-level performance: the candidate-pool factor and the relevance weight λ .

- The candidate-pool factor sets how many high-relevance tokens are considered before redundancy-aware selection is applied. For example, a candidate-pool factor of 2 means that the method first forms a candidate pool containing up to $2k_\ell$ high-relevance visual tokens at sparse layer ℓ , before selecting the final k_ℓ tokens using the relevance–redundancy rule.
- The value of λ controls the balance between text-guided relevance and redundancy reduction. A larger λ makes the selection behave more similarly to relevance-based TopK selection, while a smaller λ gives more weight to the redundancy penalty and encourages the method to select less similar tokens.

The ablation is conducted on deterministic GQA and POPE subsets of 500 samples, using the same examples across method variants. The ablation results are summarized in Table 5.7.

Table 5.7: Answer-level ablation results for redundancy-aware selection variants. GQA reports accuracy, while POPE reports accuracy and F1.

Method variant	Pool	λ	GQA (%)	Acc. POPE Acc. / F1 (%)
SparseVLM-Original	–	–	61.60	82.80 / 83.20
Ours	2	0.8	62.60	84.00 / 84.60
Ours	2	0.5	61.80	84.20 / 85.00
Ours	2	0.7	63.60	84.20 / 84.80
Ours	3	0.5	62.80	83.80 / 84.90
Ours	3	0.7	62.80	82.20 / 83.20
Threshold-Fixed	–	–	64.20	84.60 / 85.30

The ablation results suggest several findings.

Most Ours variants improve over SparseVLM-Original on the ablation subsets. On the GQA subset, SparseVLM-Original achieves 61.60% accuracy, while most Ours variants achieve higher accuracy. The strongest Ours setting on GQA is the candidate-pool factor 2 with $\lambda = 0.7$, reaching 63.60%. On the POPE subset, several Ours variants also improve over SparseVLM-Original in both accuracy and F1. This indicates that the redundancy-aware selection mechanism can provide answer-level benefits over direct SparseVLM selection under the tested ablation settings.

The best Ours setting depends on the balance between relevance and redundancy reduction. Changing λ affects how strongly the method prioritizes text-guided relevance compared with the redundancy penalty. A larger λ keeps the method closer to relevance-based selection, while a smaller λ encourages less similar selected tokens. The results show that neither extreme is automatically optimal. Instead, the best performance is obtained when relevance and redundancy reduction are balanced.

Increasing the candidate-pool factor does not automatically improve answer-level performance. A larger candidate pool gives the method more alternatives during redundancy-aware selection, but it can also introduce candidates with weaker relevance scores. In the ablation results, candidate-pool factor 3 does not consistently outperform candidate-pool factor 2. This suggests that simply giving the method more candidates is not sufficient; the candidate pool must still preserve enough task-relevant visual evidence.

Threshold-Fixed remains a strong fixed-budget baseline in this ablation. Threshold-Fixed achieves the highest GQA accuracy and the highest POPE accuracy and F1 in the reported ablation table. Therefore, the ablation results do not show that Ours is universally superior to all redundancy-aware alternatives. Instead, they show that Ours is a flexible relevance–redundancy selection method that can improve over SparseVLM-Original, while Threshold-Fixed remains a competitive hard-threshold baseline.

Overall, the ablation supports the importance of tuning the relevance–redundancy trade-off. The results suggest that redundancy-aware selection is useful, but its effectiveness depends on how the candidate pool and relevance weight are configured. Stronger redundancy reduction is not always better, and the best answer-level performance requires maintaining enough text-guided relevance while reducing unnecessary similarity among selected tokens.

5.4 Redundancy and Auxiliary Metrics

The previous section shows that the proposed method is sensitive to the balance between relevance and redundancy. This section further analyzes the selected-token behavior using auxiliary metrics, including pairwise similarity, spatial coverage, and overlap between selected patch sets. These metrics are not treated as direct performance metrics. Instead, they are used to interpret how each method selects visual tokens under sparse settings.

5.4.1 Pairwise Similarity

Pairwise similarity is used as a proxy for redundancy among selected visual tokens. A lower mean pairwise similarity suggests that the retained tokens are less similar to one another in the model’s hidden representation space. Table 5.8 reports the mean pairwise similarity of selected tokens for the Stage 8 GQA and POPE subsets.

Table 5.8: Mean pairwise similarity of selected visual tokens on Stage 8 subsets. Lower values indicate less similarity among selected token representations, but do not automatically imply better answer-level performance.

Method variant	Pool	λ	GQA mean sim.	POPE mean sim.
SparseVLM-Original	–	–	0.244	0.244
Ours	2	0.8	0.220	0.220
Ours	2	0.5	0.214	0.213
Ours	2	0.7	0.213	0.213
Ours	3	0.5	0.196	0.197
Ours	3	0.7	0.200	0.204
Threshold-Fixed	–	–	0.290	0.284

The pairwise similarity results suggest several findings.

Ours reduces average pairwise similarity compared with SparseVLM-Original.

Across both the GQA and POPE ablation subsets, all tested Ours variants produce lower mean pairwise similarity than SparseVLM-Original. SparseVLM-Original has a mean similarity of 0.244 on both subsets, while Ours reduces this value across different candidate-pool and λ settings. This is consistent with the intended behavior of the proposed method, which explicitly penalizes redundancy among selected visual tokens.

A larger candidate pool tends to reduce pairwise similarity more strongly.

The Ours variants with candidate-pool factor 3 generally achieve lower mean pairwise similarity than the variants with candidate-pool factor 2. This is expected because a larger candidate pool gives the selection rule more alternatives when searching for less redundant tokens. However, this also increases the risk of select-

ing candidates that are less relevant to the question.

Lower pairwise similarity does not automatically lead to better answer-level performance. Although the candidate-pool factor 3 variants reduce similarity more strongly, they do not consistently achieve the best GQA or POPE results in the ablation study. This shows that reducing redundancy alone is not sufficient. The selected tokens must still preserve enough text-guided relevance for the downstream question-answering task.

Threshold-Fixed achieves strong performance despite having higher pairwise similarity. Threshold-Fixed has higher mean pairwise similarity than both SparseVLM-Original and Ours in the reported subsets. Nevertheless, it achieves strong answer-level performance in several settings. This indicates that pairwise similarity should be interpreted only as a redundancy proxy, not as a direct measure of task performance.

Overall, pairwise similarity supports the mechanism of Ours but not a complete explanation of accuracy. The results show that Ours behaves as designed by selecting visual tokens that are less similar on average. However, the relationship between token diversity and answer accuracy remains indirect. Good sparse selection requires a balance between reducing redundant tokens and preserving the visual evidence needed for the question.

5.4.2 Spatial Coverage

Spatial coverage metrics provide another view of selected-token behavior. These metrics are computed from selected original patch identifiers on the 24×24 visual grid. They include quantities such as unique patch count, grid occupancy, bounding-box area, centroid distance, and quadrant coverage. Broader spatial coverage may indicate that a method preserves evidence from more regions of the image.

Ours shows spatial coverage that is broadly similar to SparseVLM-Original. Compared with SparseVLM-Original, Ours produces similar unique-patch counts and centroid distances in both GQA and POPE. Its bounding-box area is slightly higher in several nominal 128-token settings, but the differences are modest. Therefore, the spatial coverage results do not support a strong claim that Ours always selects substantially broader image regions.

Threshold-Fixed tends to produce more spatially concentrated selections. Across both datasets and both nominal sparse-budget settings, Threshold-Fixed generally has lower bounding-box area and lower centroid distance than SparseVLM-

Table 5.9: Final-layer spatial coverage of selected original visual patches. These metrics are auxiliary diagnostics and are not direct measures of answer correctness.

Dataset	Method	Budget	Unique patches	BBox area	Centroid dist.
GQA	SparseVLM-Original	64	15.44	0.778	8.09
GQA	Ours	64	15.20	0.785	8.02
GQA	Threshold-Fixed-k	64	14.99	0.641	7.00
GQA	SparseVLM-Original	128	33.70	0.901	7.93
GQA	Ours	128	32.93	0.928	8.19
GQA	Threshold-Fixed-k	128	32.51	0.850	7.54
POPE	SparseVLM-Original	64	15.44	0.806	8.31
POPE	Ours	64	15.12	0.794	8.09
POPE	Threshold-Fixed-k	64	14.97	0.651	7.20
POPE	SparseVLM-Original	128	33.88	0.939	8.32
POPE	Ours	128	33.32	0.948	8.42
POPE	Threshold-Fixed-k	128	32.88	0.882	7.82

Original and Ours. This suggests that its selected original patches are more spatially concentrated. However, this concentration should not be interpreted as automatically worse, because some questions may require evidence from a localized region.

Spatial coverage is useful as a diagnostic signal, but not as a performance metric. Broader coverage may be helpful when a question requires evidence from multiple objects or spatially separated regions. However, it may be unnecessary when the answer depends on a single object or local attribute. Therefore, spatial coverage helps describe how each method distributes selected original patches, but it does not directly explain answer-level accuracy.

Overall, the spatial coverage results provide limited but useful support for interpreting selection behavior. The table suggests that Ours does not simply collapse into a highly concentrated selection pattern, while Threshold-Fixed often selects more compact regions. These observations complement the pairwise similarity analysis, but they should be interpreted together with answer-level results, failure recovery, and qualitative visualization rather than as standalone evidence of better reasoning.

5.4.3 Selection Overlap Analysis

Selection overlap analysis compares how much the selected original patch sets coincide across different methods. In this subsection, Jaccard similarity is used as a normalized measure of overlap, making it easier to compare method pairs across datasets and sparse-budget settings.

Figure 5.2 summarizes the final-layer Jaccard similarity between selected original patch sets across method pairs. The figure is used as an auxiliary diagnostic

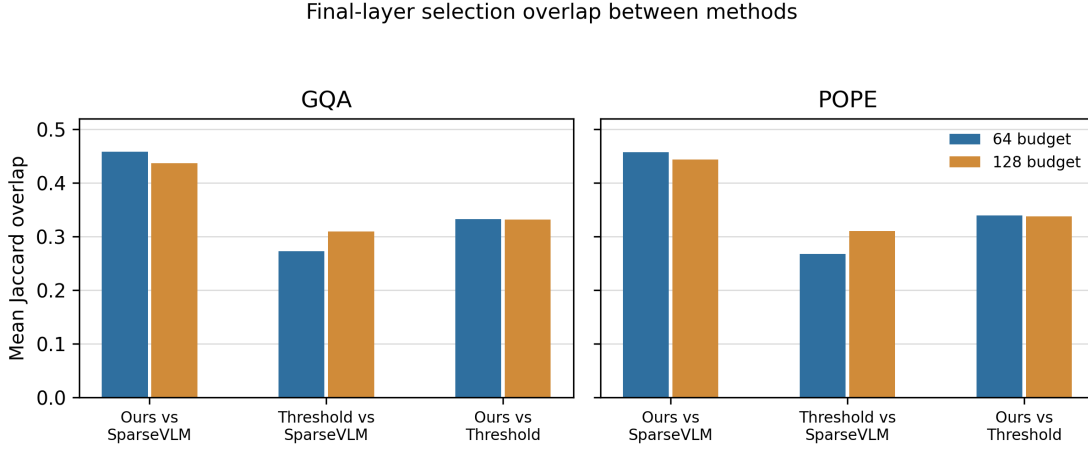


Figure 5.2: Final-layer Jaccard similarity between selected original patch sets across methods.

view of how similar or different the selected token sets are between methods.

Ours has the highest overlap with SparseVLM-Original. Across both GQA and POPE, the pair Ours versus SparseVLM-Original shows the highest Jaccard similarity among the compared method pairs. This indicates that Ours still preserves a substantial part of the relevance-based SparseVLM selection. This behavior is expected because Ours does not replace the SparseVLM text-guided relevance signal, but modifies the final selected set through redundancy-aware reranking.

Threshold-Fixed differs more strongly from SparseVLM-Original. The overlap between Threshold-Fixed and SparseVLM-Original is consistently lower than the overlap between Ours and SparseVLM-Original. This suggests that the hard-threshold filtering rule changes the selected patch set more aggressively than the MMR-style selection rule used by Ours. In other words, Threshold-Fixed does not simply behave like a small modification of SparseVLM-Original.

Ours and Threshold-Fixed are not equivalent redundancy-aware methods. The overlap between Ours and Threshold-Fixed remains moderate rather than high. Therefore, although both methods introduce redundancy awareness, they do not select the same retained visual evidence. This helps explain why the two methods may recover different failure cases or perform differently across datasets and sparse-budget settings.

The overlap pattern is consistent across GQA and POPE. The same general relationship appears in both datasets: Ours is closer to SparseVLM-Original, Threshold-Fixed is farther from SparseVLM-Original, and Ours versus Threshold-Fixed lies in between. This consistency suggests that the selection differences are

not specific to only one dataset.

Overall, the overlap analysis shows that Ours modifies SparseVLM selection without completely replacing its relevance signal. This is consistent with the design of the proposed method. Ours is intended to preserve the text-guided relevance information from SparseVLM while reducing redundancy among retained tokens. However, overlap and Jaccard similarity are still diagnostic metrics. They describe how selected patch sets differ, but they do not directly measure answer correctness or prove why a particular prediction changes.

5.4.4 Interpretation

Taken together, the auxiliary metrics support the intended mechanism of the proposed method, but only with appropriate caution. Ours reduces average pairwise similarity compared with SparseVLM-Original, which is consistent with the goal of reducing redundancy among selected visual tokens. It also changes the selected patch set rather than merely reproducing the original TopK selection.

At the same time, the auxiliary metrics show that redundancy reduction alone is not sufficient to guarantee better performance. The best selection strategy must preserve a balance between relevance and diversity. If the method prioritizes diversity too strongly, it may select tokens that are less useful for answering the question. Conversely, a method can achieve strong answer-level performance even with higher pairwise similarity if the selected tokens are highly relevant.

Therefore, the main conclusion from the auxiliary analysis is that the proposed method behaves as intended at the token-selection level, but its performance depends on maintaining the correct trade-off between text-guided relevance and redundancy reduction. These findings reinforce the need to evaluate sparse selection methods using both answer-level metrics and diagnostic token-selection metrics.

5.5 Discussion

Redundancy-aware selection and baselines. The results in this chapter show that redundancy-aware token selection can improve over direct TopK SparseVLM selection, especially under aggressive sparse settings. Across GQA, POPE, and the failure-mining diagnostic set, Ours generally performs better than SparseVLM-Original when the sparse budget becomes more constrained. This supports the central motivation of the thesis: selecting visual tokens only by individual relevance can be insufficient, because the retained set may still contain redundant information or miss evidence needed for answering the question.

At the same time, the results also show that the benefit of redundancy-aware se-

lection is not universal across all baselines and settings. Threshold-Fixed achieves stronger answer-level performance than Ours in several GQA and POPE settings. This means the thesis should not claim that the proposed method is the best sparse method overall. A more accurate interpretation is that Ours provides a flexible relevance–redundancy selection mechanism that consistently improves over SparseVLM-Original and is particularly useful for targeted failure recovery, while Threshold-Fixed remains a strong hard-threshold baseline.

The difference between broad benchmark performance and targeted failure recovery is important. On GQA and POPE, Threshold-Fixed often performs very well, suggesting that a simple hard-threshold rule can be effective when relevance-ordered candidates already contain enough useful evidence. However, on the failure-mining diagnostic set, Ours shows a clearer advantage under the aggressive sparse setting by recovering more SparseVLM-Original failures and introducing fewer regressions than Threshold-Fixed. This suggests that the continuous relevance–redundancy trade-off may be more useful in cases where the selection boundary is difficult and a binary threshold is too rigid.

Interpretation of diagnostic evidence. The auxiliary metrics help explain the behavior of the selection methods, but they must be interpreted carefully. Pairwise similarity results show that Ours reduces average similarity among selected visual tokens compared with SparseVLM-Original. This is consistent with the intended design of the method. However, lower similarity does not automatically lead to better task performance. The ablation results show that overly emphasizing diversity, or expanding the candidate pool too much, can reduce answer-level performance. Therefore, the main issue is not simply to minimize redundancy, but to preserve a useful balance between text-guided relevance and token diversity.

The qualitative visualization analysis provides additional diagnostic support, but it does not prove the causal mechanism. Some recovered cases appear consistent with the interpretation that Ours preserves more answer-relevant evidence than SparseVLM-Original. However, the visual evidence is not consistently clear across all cases, and recycled or compressed tokens cannot be directly visualized as original image patches. Therefore, visualization should be treated as a tool for interpretation rather than proof that a specific selected patch caused an answer change.

Main interpretation. Overall, the findings support a careful and limited claim. The proposed method improves over SparseVLM-Original by introducing redun-

dancy awareness into visual-token selection. Its strongest evidence appears under aggressive pruning and targeted failure analysis, where preserving a more diverse but still relevant set of visual tokens can help recover some SparseVLM failures. However, the method is not universally superior to all alternatives, and redundancy reduction alone is not sufficient to guarantee better performance. The effectiveness of sparse visual-token selection depends on balancing relevance, redundancy, token budget, and downstream reasoning requirements.

Implications for future work. These findings motivate several directions for future improvement:

1. **Adaptive relevance–redundancy trade-off.** The relevance–redundancy balance could be adjusted dynamically based on question type, image complexity, sparse layer, or model uncertainty, instead of using a fixed setting for all examples.
2. **Hybrid selection methods.** Threshold-based and MMR-style selection could be combined to benefit from both interpretable hard filtering and continuous relevance–redundancy scoring.
3. **Broader benchmark evaluation.** Future work should evaluate the method on larger and more diverse benchmark settings to better understand when redundancy-aware selection helps most.

These points are discussed further in the conclusion and future work chapter.

CHAPTER 6. Conclusion and Future Work

6.1 Conclusion

This thesis studied efficient visual-token selection for image-based VLMs, focusing on redundancy in SparseVLM-style sparse inference. Direct TopK selection can retain tokens that are individually relevant to the prompt but mutually similar. Under aggressive token budgets, such redundancy may reduce the capacity available for other useful evidence, including small objects, attributes, relations, and localized details.

To address this issue, the thesis proposed a redundancy-aware visual-token selection method. The method first considers a high-relevance candidate pool and then selects tokens by balancing text-guided relevance with a redundancy penalty against already selected tokens. This design preserves the relevance signal used by SparseVLM while adding an explicit mechanism for encouraging diversity among retained visual tokens.

The experiments show that the proposed method improves over SparseVLM-Original in several important settings. On GQA, Ours improves accuracy at both the 128-token and 64-token budgets. On POPE, Ours improves both accuracy and F1 under the aggressive 64-token setting, where preserving object-level evidence is especially important. The strongest evidence appears in the failure-mining diagnostic set: Ours improves correctness from 70.00% to 75.00%, with 10 recoveries and 5 regressions.

The auxiliary analysis further supports the intended behavior of the method. Ours reduces average pairwise similarity among selected tokens compared with SparseVLM-Original, indicating that the selected set becomes less redundant according to this proxy. However, the results also show that redundancy reduction alone is not sufficient. The best sparse-selection behavior depends on maintaining a useful balance between prompt relevance, token diversity, budget constraints, and downstream reasoning requirements.

6.2 Limitations

The findings should be interpreted with several limitations. First, the proposed method is not universally superior to all sparse baselines. Threshold-Fixed remains stronger than Ours in several broad GQA and POPE settings, especially when a simple hard-threshold rule retains enough useful evidence. Therefore, the contribution of this thesis should be understood as a redundancy-aware improvement

over SparseVLM-Original, not as a method that always dominates threshold-based selection.

Second, the failure-mining set is intentionally diagnostic. It is useful for analyzing targeted SparseVLM-Original failures, but it does not replace large-scale benchmark evaluation. Third, visualization provides qualitative support only. Retained original patches can suggest how different methods select visual evidence, but recycled or compressed tokens cannot be directly visualized as original image regions, and the visualization alone cannot prove causal explanations for answer changes. Finally, auxiliary metrics such as pairwise similarity and spatial coverage are interpretive signals rather than direct measures of VLM reasoning quality.

6.3 Future Work

Several directions can extend this work.

1. **Adaptive relevance–redundancy balancing.** Future work could make the relevance–redundancy trade-off adaptive instead of using a fixed setting. The selection behavior could be adjusted according to question type, image complexity, sparse layer, token budget, or model uncertainty. This may help the method preserve more task-relevant evidence while still reducing redundant tokens.
2. **Hybrid redundancy-aware selection.** The results show that Ours and Threshold-Fixed have different strengths. Ours performs well in targeted failure recovery, while Threshold-Fixed remains strong in several benchmark settings. Future work could combine MMR-style continuous scoring with threshold-based filtering to benefit from both approaches.
3. **Larger and more diverse evaluation.** Future work should evaluate redundancy-aware selection on larger benchmark subsets and additional multimodal tasks. This may include more diverse VQA datasets, hallucination benchmarks, OCR-based VQA, fine-grained reasoning tasks, and video-language settings. Such evaluation would help clarify when redundancy-aware selection is most effective.
4. **Better interpretability and efficiency analysis.** Future studies should develop better tools to explain how selected patches, recycled tokens, and compressed representations affect final predictions. In addition, a more complete efficiency analysis could include memory usage, throughput, GPU utilization, and hardware-specific inference cost, beyond the per-sample latency reported in this thesis.

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